

REPLACEMENT / MAINTENANCE

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SECTION 1 - RAMPDOWN TO ZERO FIELD



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.
- FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.
- HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.



THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING THE MAGNET:

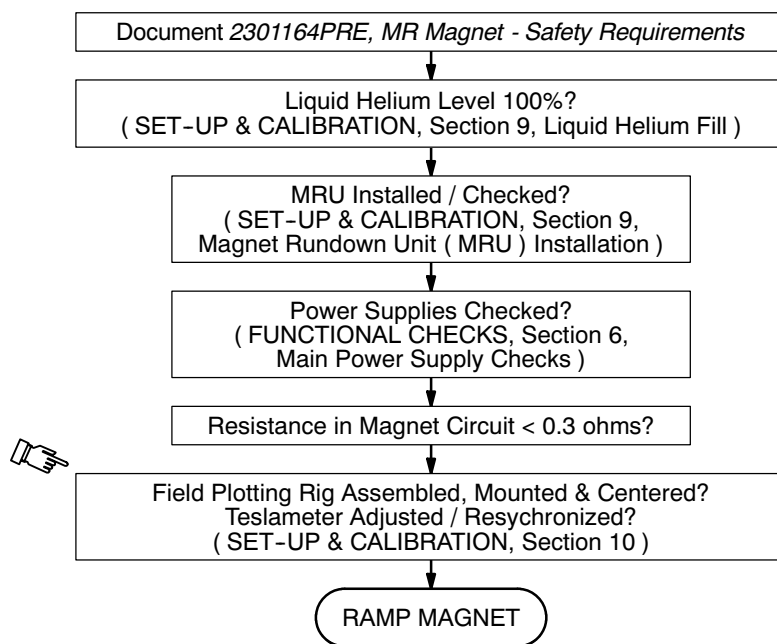
- HELIUM GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:
 - A. MAKE SURE THAT THE MAGNET ROOM DOOR IS PROPPED OPEN,
 - B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND
 - C. REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, INCLUDED WITH THIS MANUAL.

SECTION 1 - RAMPDOWN TO ZERO FIELD (continued)

**WARNING!**

(CONTINUED)

- NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN.
- POST WARNING SIGNS (46-255326P1) OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS AND OTHER BIOSTIMULATION DEVICES NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.
- POST “SECURITY ZONE” (46-255325P1) AND “AUTHORIZED PERSONNEL ONLY” (2289812) SIGNS AT THE MAGNET ROOM ENTRANCE AND MR SUITE ENTRANCE, RESPECTIVELY, PRIOR TO RAMPING THE MAGNET. THESE WARNINGS ARE TO ALERT PERSONNEL THAT NO FERROMAGNETIC MATERIAL OR INDIVIDUALS WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES ARE ALLOWED IN THE MAGNET ROOM WHEN THE MAGNET IS RAMPED. SEE 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.
- REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. PULL THE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND SITE GEOMETRY ALLOW. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.
- MAKE SURE THE MAGNET RUNDOWN UNIT (MRU) IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE SET-UP AND CALIBRATION, SECTION 4.
- MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT A PROPERLY FUNCTIONING OXYGEN MONITOR IS USED IN THE MAGNET AND/OR CRYOGEN STORAGE ROOM(S) BY SERVICE PERSONNEL.
- MAKE SURE THE MAGNET IS AT LEAST 90% (≥ 1040 mm) FULL OF LIQUID HELIUM TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING DURING RAMPING TO A POINT WHERE A QUENCH MAY OCCUR.

SECTION 1 - RAMPDOWN TO ZERO FIELD (continued)

RAMPING FLOW DIAGRAM
ILLUSTRATION 1-1

1-1 TOOLS / EQUIPMENT

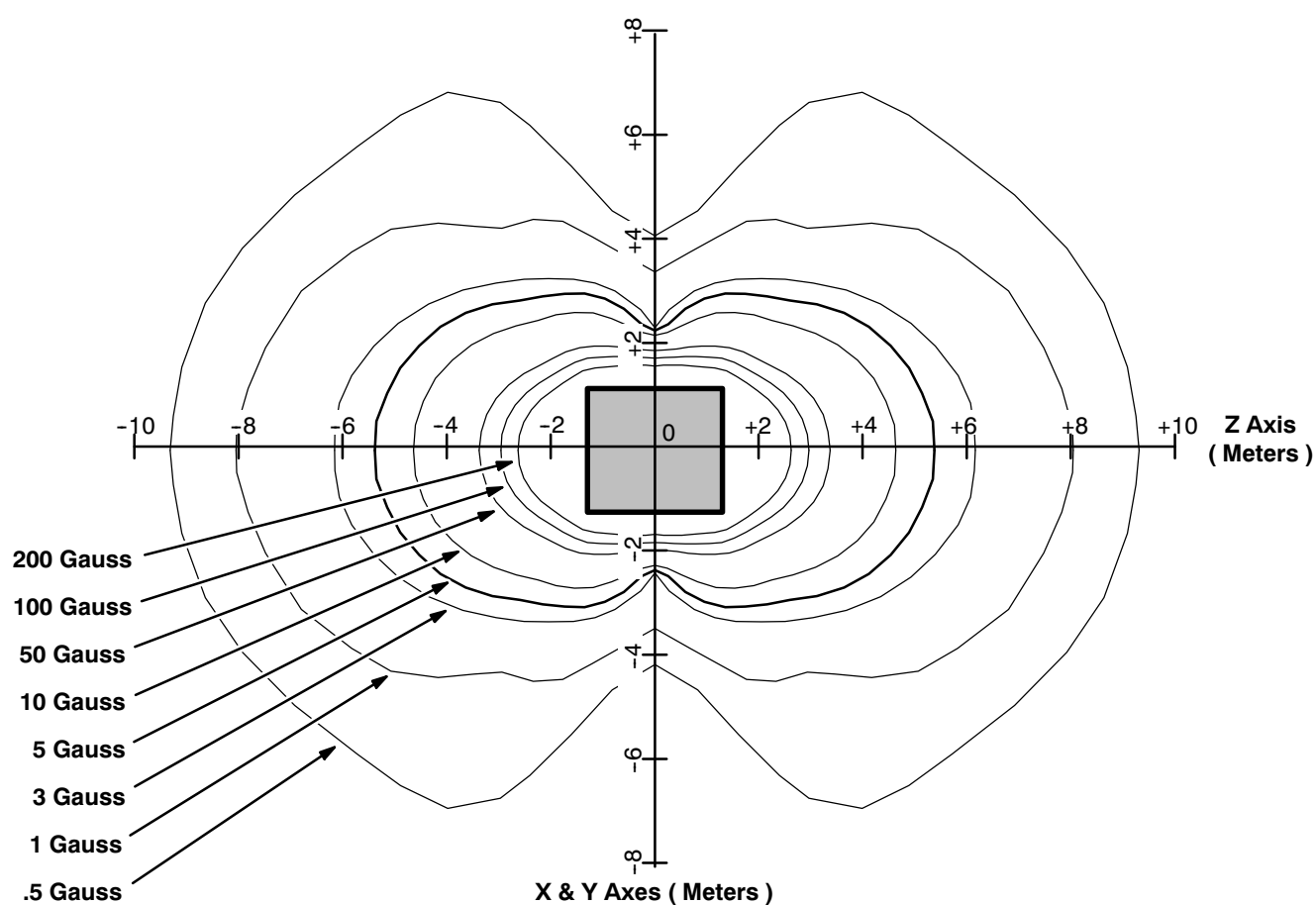
- Ramp / Energize Kit (2296056)
- Shimming Kit (2297304)
- Shim Power Supply (46-260777G3)
- Ramp Power Supply. The 1000 amp ESS7.5-1000-2-D-1236 (46-260776G4) is recommended because of its reduced ripple.
- Shim Cable Kit (2135558)
- Ramp Cable Kit (2135435)
- Metrolab Teslameter Kit (46-251865G4)
- No. 6 MetroLab Field Probe (2295525-5)
- Main Switch Adapter Box (2320946)

1-2 PREPARATION

1. Make sure the liquid helium level inside the magnet is $\geq 90\%$ (1040 mm). Top off the magnet in accordance with REPLACEMENT / MAINTENANCE, Section 3, if not.
2. Position the power supplies outside the magnet's 5 gauss zone (extending from the magnet's center approximately 11 feet or 3.3 meters on either side of the magnet and 19.5 feet or 6 meters at either end). See Illustration 1-2.

IMPORTANT !!!

DO NOT BLOCK the magnet room entrance or any other passageway.



GEMSO 3T/940 FRINGE FIELD
ILLUSTRATION 1-2

3. Perform the power supply set-up checks detailed in FUNCTIONAL CHECKS, Section 6-2-4.
4. Assemble, mount and center the Field Plotting Rig (2292542) in conformance with SET-UP AND CALIBRATION, Section 10-1, Field Plotting Rig (2292542) Set-Up.

1-2 PREPARATION (continued)

5. Remove the Pressure Gauge Manifold and install the Exhaust Gas Vent Assembly (2292536) in its place. See Illustration 1-3 for the factory-installed configuration and Illustration 1-4 for the Exhaust Gas Vent arrangement.

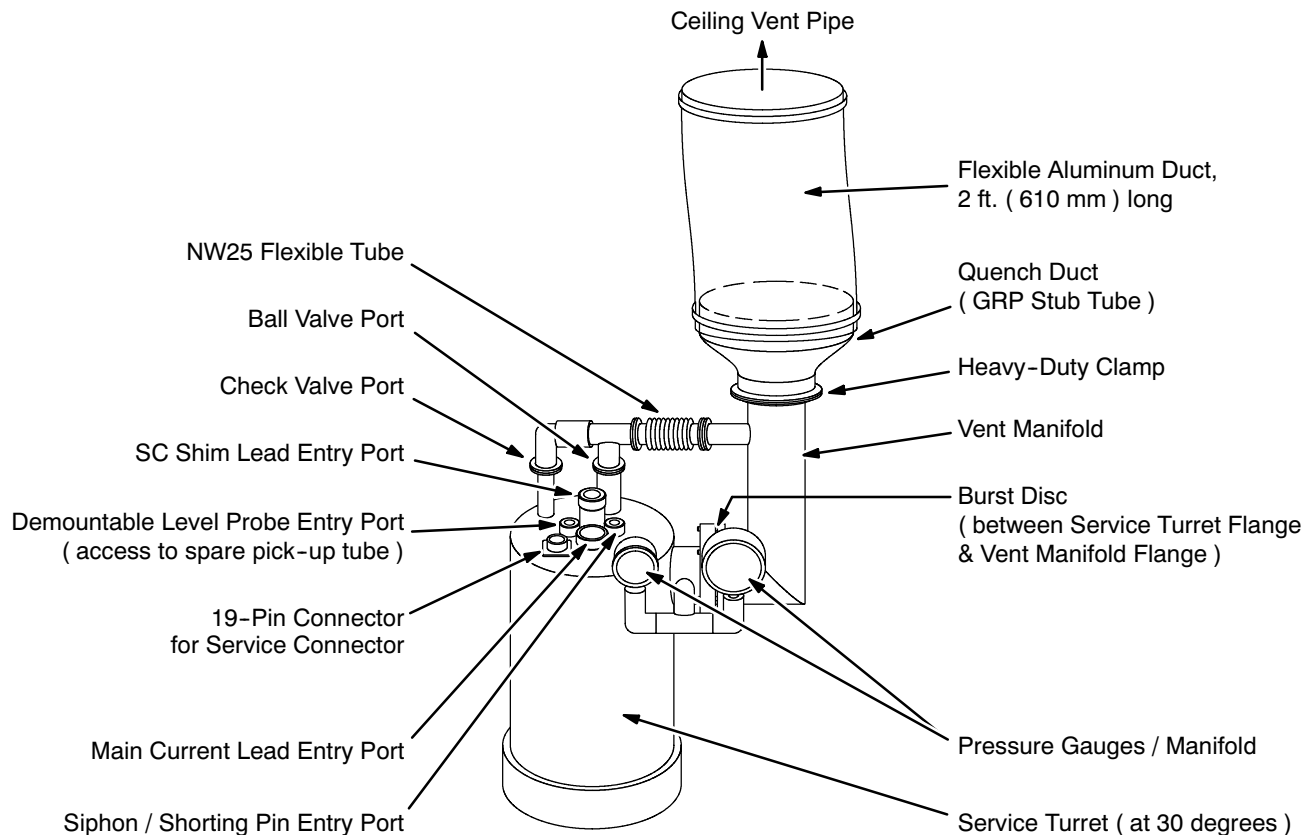
**TYPICAL FACTORY-INSTALLED 3T SERVICE TURRET PLUMBING**

ILLUSTRATION 1-3

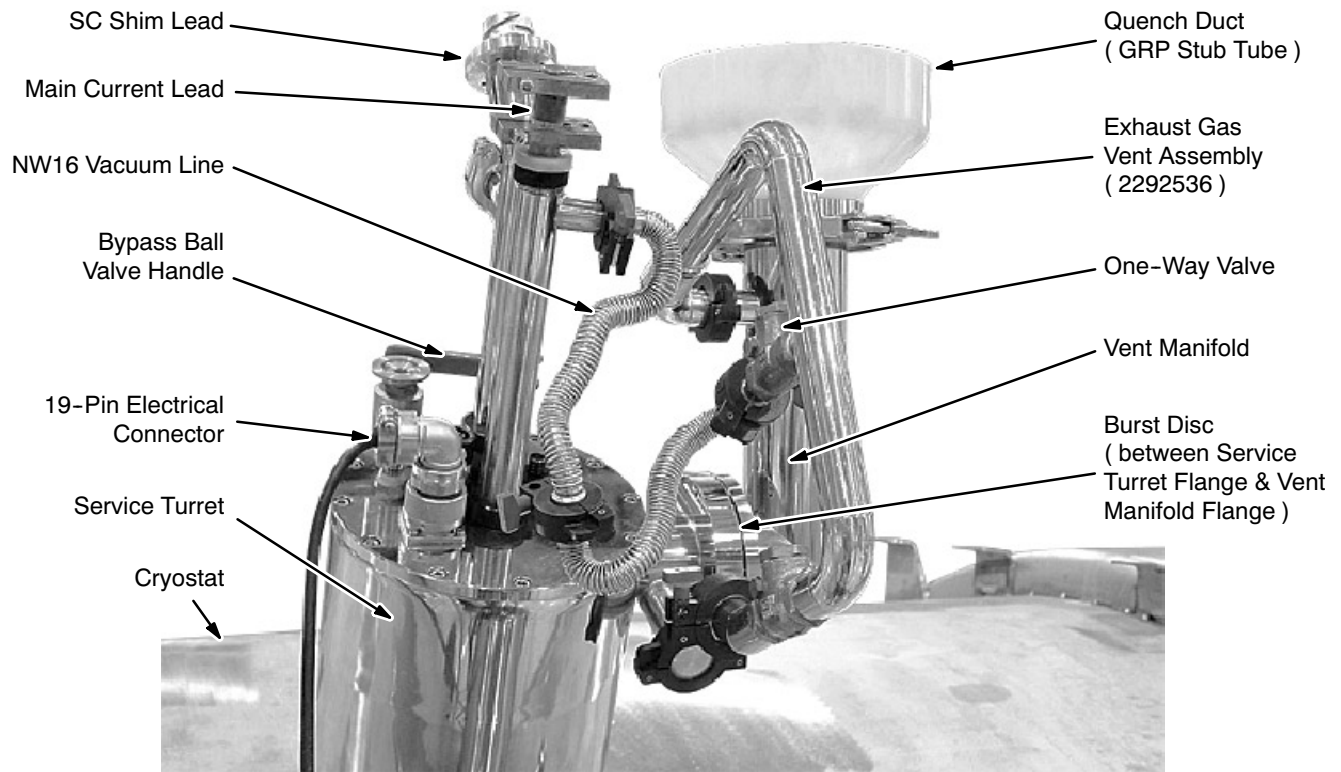
1-2 PREPARATION (continued)**EXHAUST GAS VENT CONNECTION**

ILLUSTRATION 1-4

6. Make sure the terminals on the Main Current Lead are not shorting to each other or the other end of the current lead.
7. Connect the Main Ramp Cables to the copper lugs on the Main Current Lead. See Illustration 1-5. The positive current lead terminal is the center top block and has a M12 connection screw. The negative terminal is the lower block and has a M10 screw.

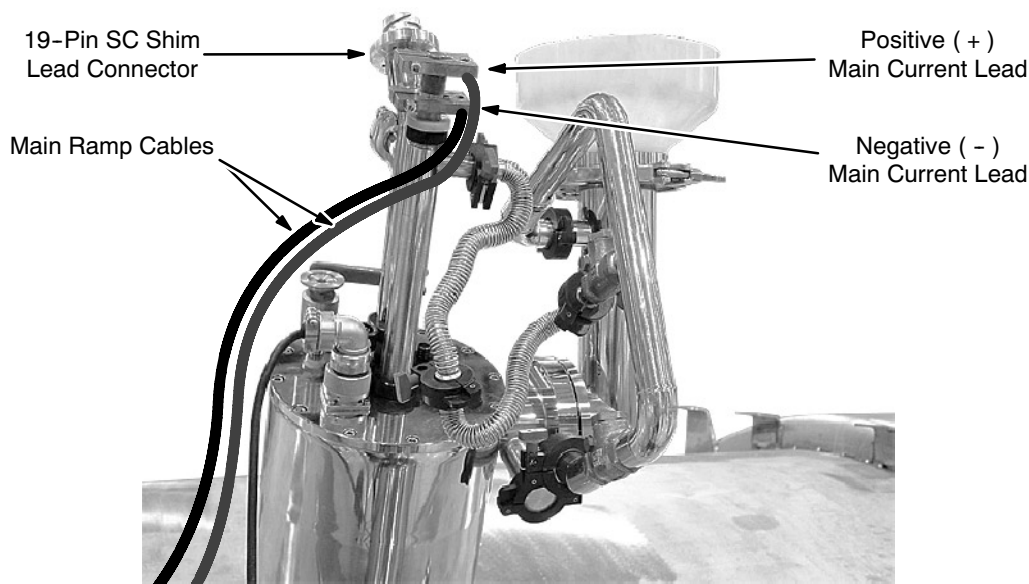
1-2 PREPARATION (continued)**SHIM AND CURRENT LEAD CONNECTIONS**

ILLUSTRATION 1-5

8. Connect the Shim Heater Power Supply to the Shim Adapter Box (2320885). See Illustrations 1-6 and 1-8.
9. Connect the Main Shim Cable (46-260726G1) between Shim Power Supply socket J2 and the input socket on the Shim Adapter Box. See Illustrations 1-6, 1-7 and 1-8.

CAUTION

To prevent Shim Heater Coil burnout, DO NOT connect the heater supply from the Shim Power Supply. Only use the separate Shim Heater Power Supply.

10. Connect Shim Cable 2297280 to the Shim Adapter Box's 23-pin output and then to the Shim Lead. See Illustrations 1-6 and 1-8.

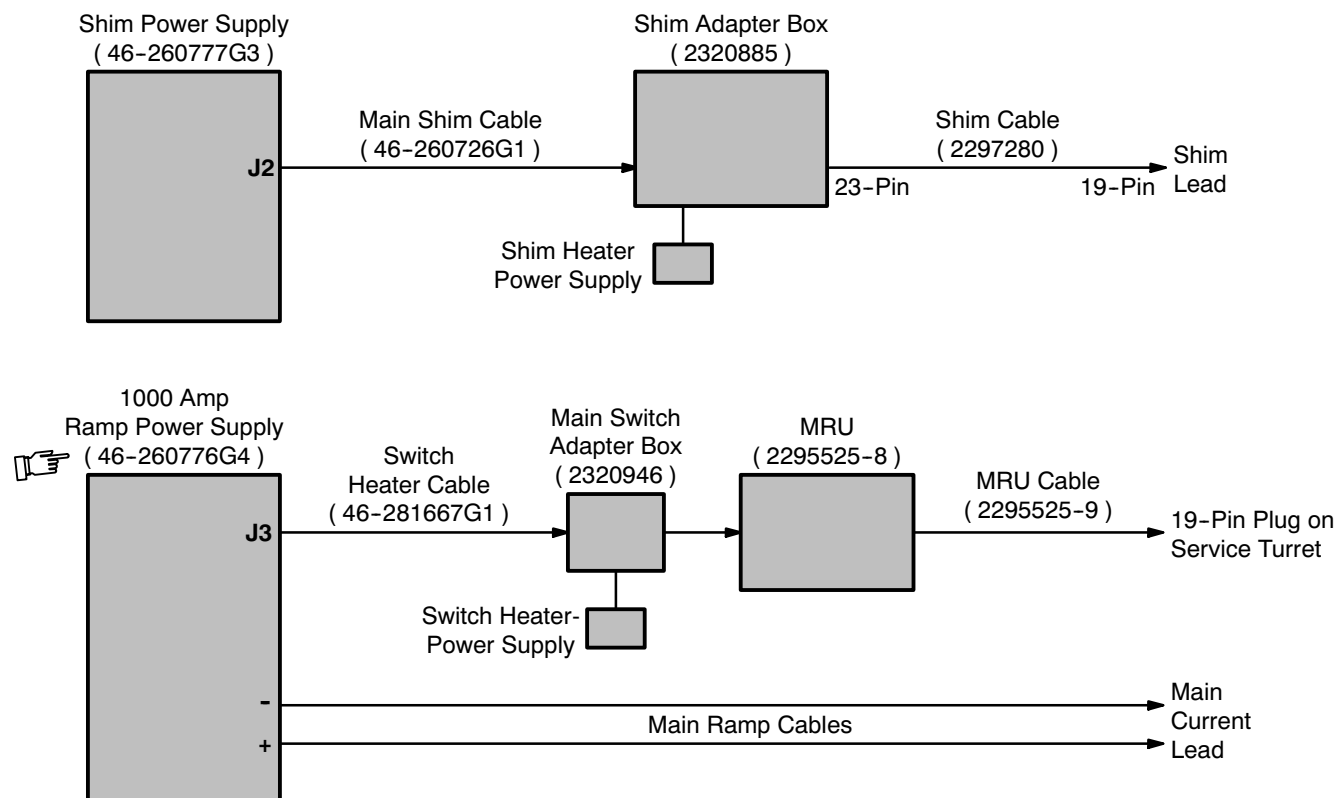
1-2 PREPARATION (continued)**CABLE CONNECTIONS FOR RAMPDOWN**

ILLUSTRATION 1-6

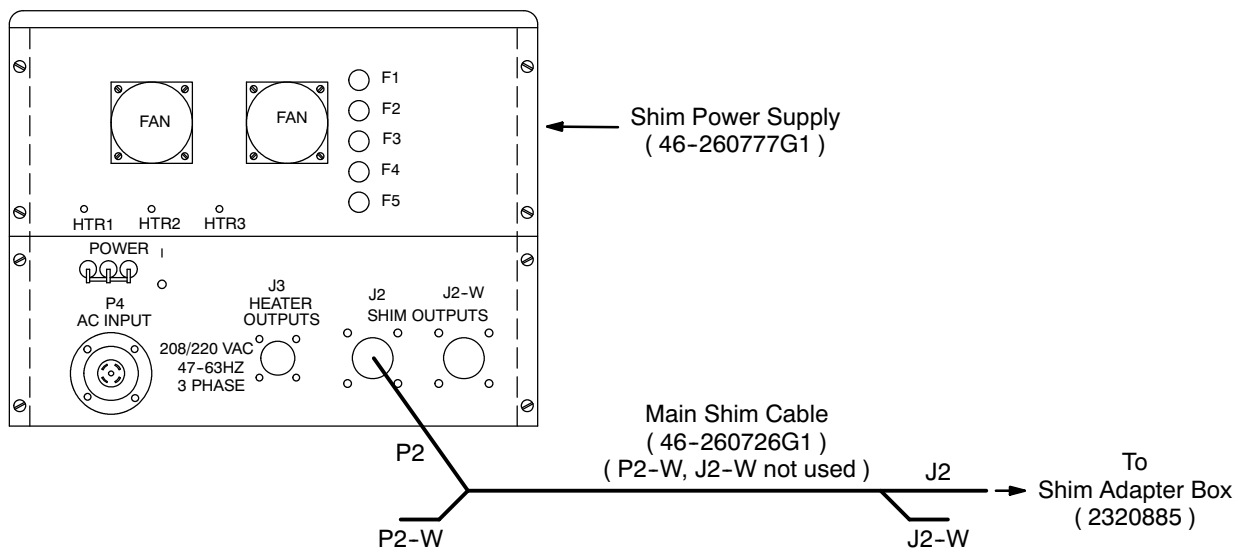
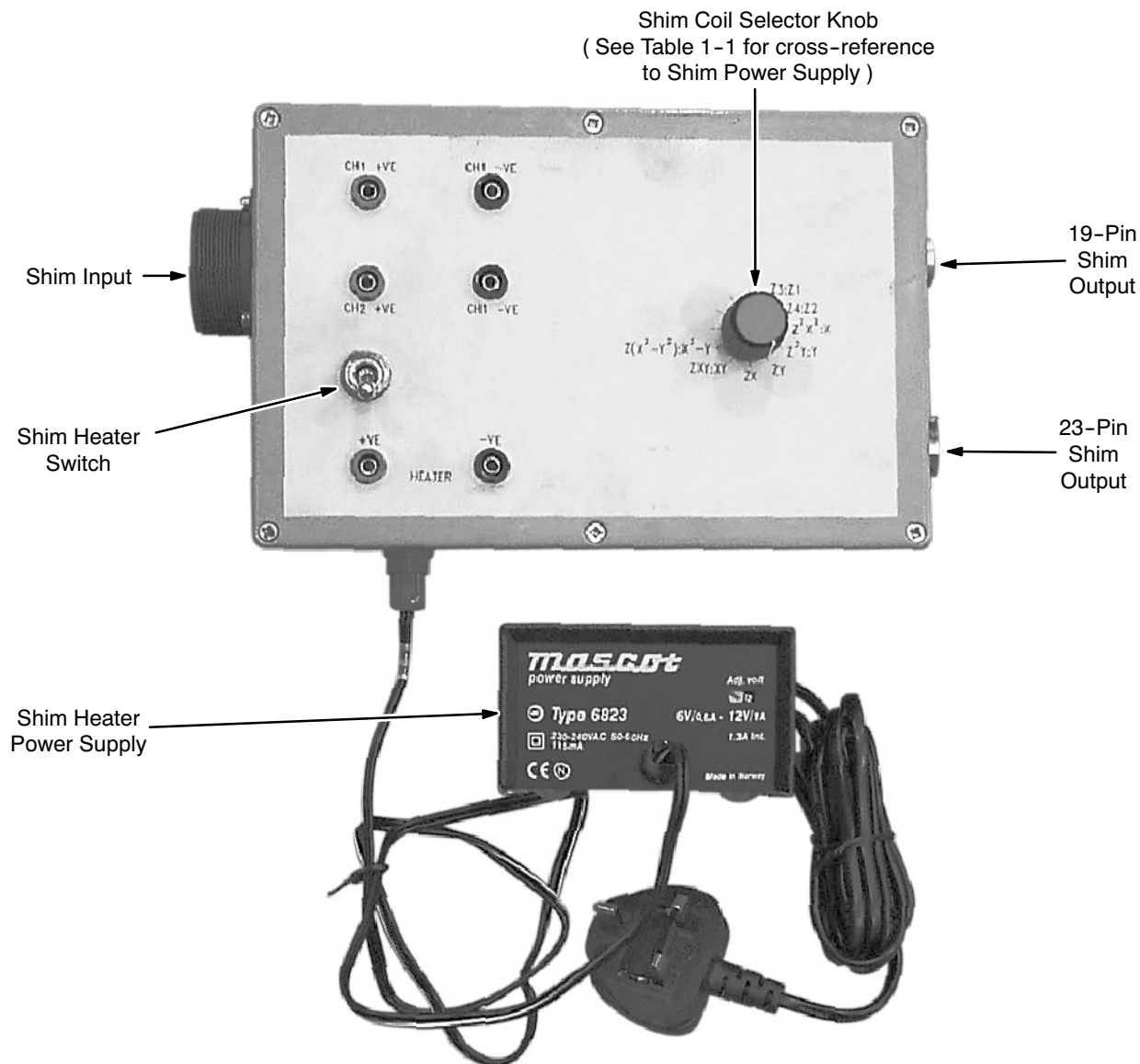
**SHIM POWER SUPPLY OUTPUT CONNECTIONS**

ILLUSTRATION 1-7

1-2 PREPARATION (continued)



SHIM ADAPTER BOX (2320885)

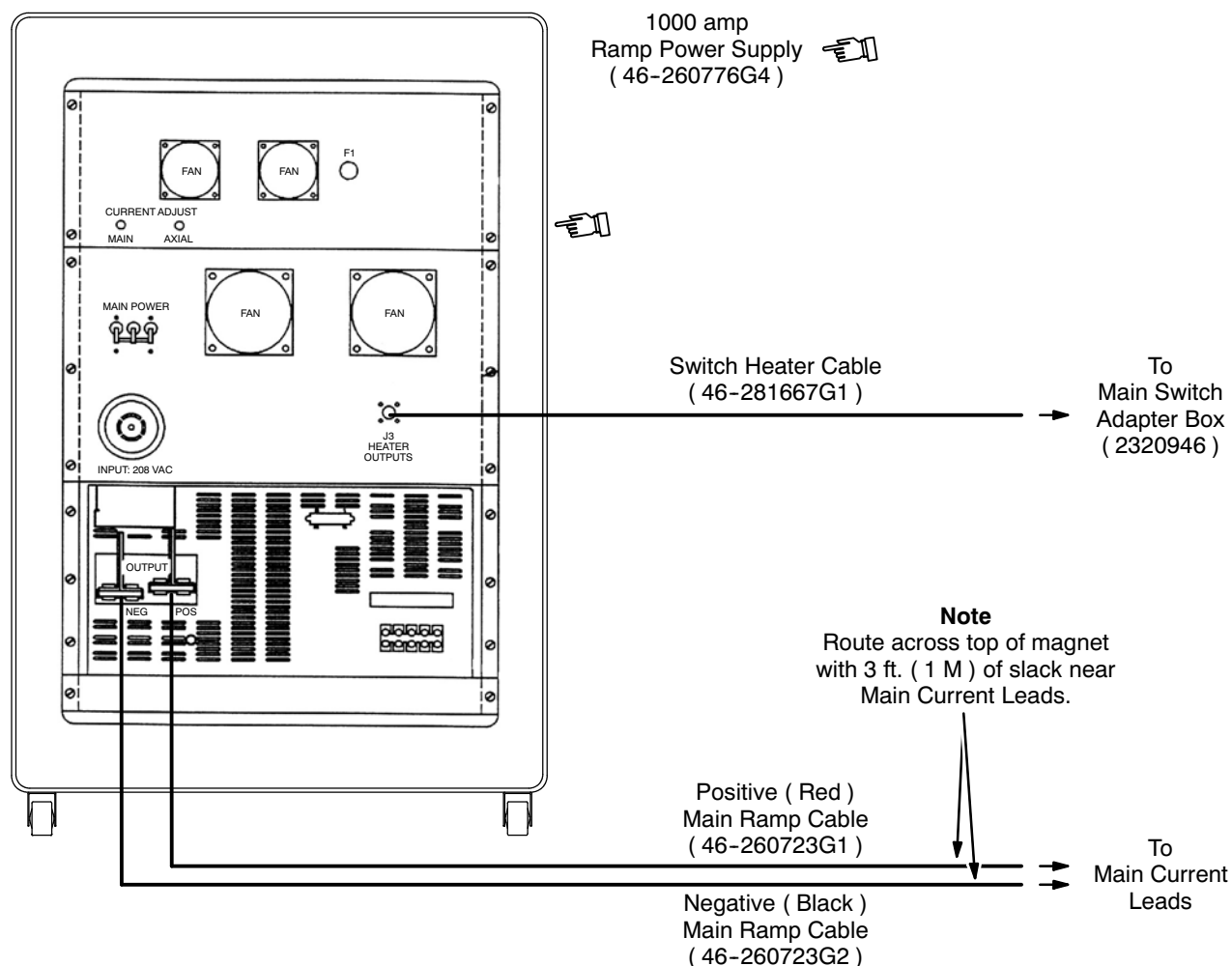
ILLUSTRATION 1-8

TABLE 1-1
CROSS-REFERENCE, SHIM POWER SUPPLY CHANNELS TO SHIM ORDERS

Power Supply Channel (Shim Power Supply)	T1-1						T1-2							
Shim Order (Shim Adapter Box Label)	Z3	Z4	Z ² X	Z ² Y	ZXY	Z(X ² -Y ²)	Z1	Z2	X	Y	ZX	ZY	XY	X ² -Y ²

1-2 PREPARATION (continued)

11. Connect the Switch Heater Cable (46-281667G1) to the Ramp Power Supply's connector J3 and then to the Main Switch Adapter Box (2320946). See Illustrations 1-6, 1-9 and 1-10.
12. Connect a digital voltmeter (DVM), set to its volts range, to the test points on the Main Switch Adapter Box.
13. Plug in the Switch Heater Power Supply. Verify that the Main Magnet Switch on the Main Switch Adapter Box opens (on) and closes (off); leave the switch in the closed position. Unplug the Switch Heater Power Supply.



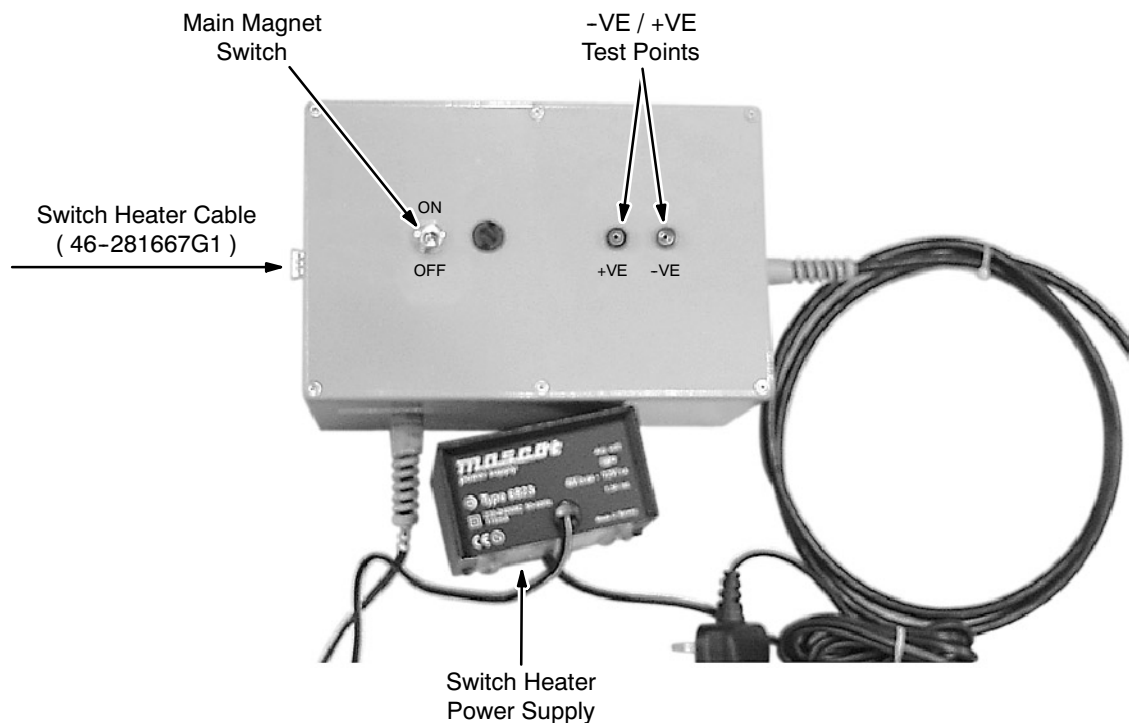
MAGNET POWER SUPPLY CONNECTIONS (1000 AMP)
ILLUSTRATION 1-9

1-2 PREPARATION (continued)

14. Connect the Main Switch Adapter Box to the MRU's 9-pin female connector labeled "PSU." See Illustrations 1-6 and 1-10.

Note

This arrangement gives access to the main magnet voltage and allows the magnet switch to be opened with the separate Switch Heater Power Supply.



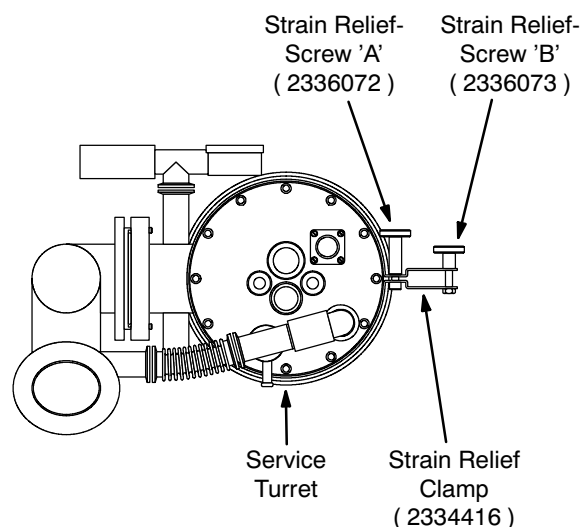
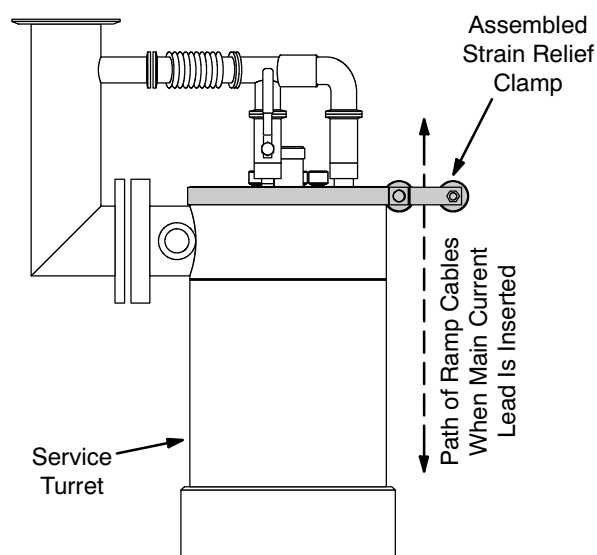
MAIN SWITCH ADAPTER BOX (2320946)
ILLUSTRATION 1-10

15. Install the Ramp Cable Strain Relief Clamp (Clamp 22334416, Screw 'A' 2336072 and Screw 'B' 2336073) around the top edge of the Service Turret as shown in Illustration 1-11, if not already installed.



Do not leave the Main Current Lead Port open to the atmosphere for more than a few seconds since air will condense in the port, which will cause an ice blockage.

16. Remove the cap from the Main Current Entry Port, moving the nut, washers and o-ring to the Main Current Lead. See Illustration 1-3.

1-2 PREPARATION (continued)**Top View of Service Turret****Side View of Service Turret****RAMP CABLE STRAIN RELIEF CLAMP
ILLUSTRATION 1-11****WARNING!**

MAKE SURE THE MAIN CURRENT LEAD / SHIM LEAD EXHAUST PORTS ALWAYS FACE AWAY FROM THE SERVICE PLATFORM TO PREVENT COLD HELIUM GAS EXHAUSTING TOWARD THE SERVICE ENGINEER.

17. Insert the Main Current Lead into the Main Current Entry Port and push down gradually. Cold helium gas will emerge from the entry port. If the rush of gas becomes too great, hold the lead steady or retract it slightly until the rush decreases.
18. Once contact is felt, push the Main Current Lead into full engagement with the male connector at the bottom of the entry port.

Note

Contact is necessary over the whole 2 inches (50 mm) of the male connector's pin to minimize contact resistance.

1-2 PREPARATION (continued)

Do not force the Main Current Lead into full engagement.

19. Rotate the Main Current Lead gently to its fully engaged position.

IMPORTANT !!!

Do not allow the cables or exhaust port touch the Cryostat. Support the cables via a strain relief to prevent the cables' weight from pulling on the Current Lead.

20. Tighten the nut to create a gas-tight seal.
21. Run the Ramp Cables through the Ramp Cable Strain Relief Clamp along the path shown in Illustration 1-11. Make sure the cables are not pulling on the Main Current Leads, then tighten the clamp.
22. Remove the cap from the SC Shim Lead Entry Port, moving the nut, washers and o-ring to the Shim Lead. See Illustration 11-3.



MAKE SURE THE MAIN CURRENT LEAD / SHIM LEAD EXHAUST PORTS ALWAYS FACE AWAY FROM THE SERVICE PLATFORM TO PREVENT COLD HELIUM GAS EXHAUSTING TOWARD THE SERVICE ENGINEER.

23. Insert the Shim Lead into the SC Shim Lead Entry Port and push down gradually. Cold helium gas will emerge from the entry port. If the rush of gas becomes too great, hold the lead steady or retract it slightly until the rush decreases.

1-2 PREPARATION (continued)

24. Once contact is felt, push the Shim Lead into full engagement.

Note

The Shim Lead is keyed, so it should only engage in one orientation: with the exhaust port facing directly away from the service platform. If it does not fit in the correct orientation, stop immediately; **DO NOT FORCE THE SHIM LEAD INTO ENGAGEMENT NOR ROTATE IT.** Instead, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715).

IMPORTANT !!!

If ice is encountered on the Shim Lead contacts, remove and warm the Shim Lead, then reinsert the warm lead to clear the ice.

25. Connecting the 19-Pin Breakout Box (2295141) to the lead and measuring the resistances between pin "J" and the other pins as listed in Table 1-2. If all measurements are within the specified ranges, the Shim Lead has been engaged correctly. If any measurement is outside its expected range, contact GE Medical Systems Oxford (8-011-44-1235-5490-00) or Florence (Final Test: 843-664-1715).

TABLE 1-2
RESISTANCE CHECKING OF CORRECT SHIM LEAD INSERTION ORIENTATION

Pin Pair (at 19-Pin Breakout Box)	A - J	B - J	C - J	D - J	E - J	F - J	G - J	H - J
Expected Resistance Measurement (Ohms)*	200 ± 20	200 ± 20	200 ± 20	200 ± 20	100 ± 10	100 ± 10	200 ± 20	200 ± 20

26. Measure the resistance between the Main Current Lead's positive and negative terminals. Clean and reseal the Main Current Lead if resistance is not < 0.3 ohms.
27. Connect a NW16 vacuum line between the Main Current Lead exhaust and the one-way valve at the center of the Exhaust Gas Vent Assembly (2292536) to route exhaust gas safely out of the room through the quench duct. See Illustration 1-4.
28. Blank off the Shim Lead exhaust to make sure all exhaust gas is used to cool the Current Lead while ramping down the magnet.
29. Connect the ramp cables to the power supply. Check the polarity.
30. Adjust / resynchronize the Teslameter in conformance with SET-UP AND CALIBRATION, Section 10-2.

1-3 RAMPING DOWN TO ZERO FIELD

1. Turn on the Ramp Power Supply.
2. On the MRU turn on the B0 Heater.
3. Turn on the Ramp Power Supply's Axial Switch Heater.
4. Turn on the Shim Power Supply.
5. Remove the Shorting Pin:
 - a. Remove the plug from the Siphon / Shorting Pin Entry Port.
 - b. Slowly insert the Shorting Pin Insertion Tool until it contacts the Shorting Pin.
 - c. Push the tool down and rotate it 10 turns clockwise.
 - d. Once the Shorting Pin is fully engaged, pull the Shorting Pin Insertion Tool out.
 - e. Replug the Siphon / Shorting Pin Entry Port.
6. Make sure all current potentiometers on the Shim Power Supply are set to zero (fully counterclockwise).
7. Set all shims to zero current:
 - a. Select the "Z1:Z3" shim pair via the Shim Coil Selector Knob on the Shim Adapter Box.
 - b. If the magnet has been shimmed previously, set the current and polarity in the T1 - 1 and T1 - 2 channels equal to that in the shims. Refer to SET-UP AND CALIBRATION, Section 12-4-3, Steps 2 through 13, although it is not necessary to position the probe or wait for the field to stabilize.
 - c. Turn on the Shim Heater Switch, then wait 10 seconds for the switch to go resistive.
 - d. Reduce the currents to zero, then wait one minute before turning off the Shim Heater Switch.
 - e. Select the next shim pair. Repeat Steps 7b - 7d until all shims have been set to zero.
8. Select the "Z2:Z4" shim pair and turn on the Shim Heater Switch to hold these open during the ramp down.

1-3 RAMPING DOWN TO ZERO FIELD (continued)**WARNING!**

THE MAXIMUM CURRENT LIMIT SETTING MUST NEVER EXCEED 260 AMPS TO PREVENT POTENTIAL MAIN COIL BURN-OUT.

WARNING!

THE POWER SUPPLY MUST BE SET TO THE SAME POLARITY AND CURRENT AS THE LAST VALUES RECORDED IN DATA SHEETS, SECTION 7, TO PREVENT A MAGNET QUENCH WHEN TURNING ON THE MAIN MAGNET SWITCH.

9. Set the Current Limit on the Ramp Power Supply to one amp more than the persistence current value recorded under "Final Persistence Current (A)" in DATA SHEETS, Section 7, Table 7-1.
10. Set the magnet voltage to zero and then increase the voltage until the magnet current matches the recorded persistence current value.
11. Plug in the Switch Heater Power Supply on the Main Switch Adapter Box and wait 5 minutes for it to go resistive. If the field begins drifting upwards during this period, slightly reduce the voltage until the field starts reducing.
12. Once the switch is resistive, decrease the voltage until the field begins to reduce.
13. Continue periodically decreasing the voltage in the stages shown in Table 1-3 in order to keep the field running down. Keep the voltage within 100 mV of the "Voltage" value in Table 1-3 (e.g., 0.4 – 0.5 volts for full field to 249.0 amps).

TABLE 1-3
RECOMMENDED MAXIMUM ENERGIZATION RATES

Main Coil Current (Amps)	Current Limit (Amps)	Amps / Hour	Main Coil Voltage*	Duration
Final Persistence Value* – 249.0	Final Persistence Value* + 1.0	4.4	0.5	1.0 hours
249.0 – 237.0	250	8.7	1.0	1.4 Hours
237.0 – 186.0	238	17.4	2.0	2.9 Hours
186.0 – 0.0	187	21.8	2.5	8.5 Hours
* "Final Persistence Current (A)" value from DATA SHEETS, Section 7, Table 7-1.				

1-3 RAMPING DOWN TO ZERO FIELD (continued)

14. When the current reaches zero, adjust both voltage and current to zero and wait for magnet voltage to decay to zero ($\approx$ 10 minutes).
15. Turn off the Main Magnet Switch on the Main Switch Adapter Box.
16. Turn off the Shim Heater Switch and the B0 Coil Heater Switch.
17. Remove the Shim Lead and the Main Current Lead.
18. Remove the Exhaust Gas Vent Assembly and reinstall the Pressure Gauge Manifold.

SECTION 2 - BURST DISC REPLACEMENT



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

2-1 INTRODUCTION

Defective Burst Discs and those that rupture during a magnet quench must be replaced to prevent cryopumping and ice blockages forming in the system. Quenches release cryogenics, so cryogen refill will be required. If cryogen refill cannot take place during the Burst Disc replacement service call, apply helium gas to the Helium Vessel to prevent continued cryopumping and ice block formation in the Exhaust Lines and Vertical Stack.

2-2 TOOLS / EQUIPMENT

- Replacement Burst Disc (2255516)
- Portable Oxygen Monitor (2106236 or 2106237) if Oxygen Monitor Kit 2107184 is not installed and functional in room.
- Multimeter
- 5 mm Allen Key

2-3 VERIFYING BURST DISC REPLACEMENT IS REQUIRED

1. Check the Service Turret Pressure Gauge. A broken Burst Disc doesn't hold pressure in the Cryostat.
2. Connect common and Ohms leads from a Multimeter to the Burst Disc's "Tell" leads; the meter should read "O.L." Ohms if the disc is broken.

Note

Remove and visually inspect the Burst Disc if not absolutely certain its condition is good.

2-4 BURST DISC REMOVAL

WARNING!

MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, INCLUDED WITH THIS MANUAL.

WEAR PROTECTIVE CLOTHING (LONG-SLEEVE SHIRT, LONG PANTS AND A PROTECTIVE APRON / JACKET), INSULATED NONABSORBENT GLOVES AND GOGGLES / SAFETY GLASSES WHEN REPLACING THE BURST DISC ON COLD VENT SYSTEM.

MAKE SURE THE MAGNET IS RAMPED DOWN TO ZERO FIELD TO PREVENT ANY POSSIBILITY OF A QUENCH.

VENT OFF CRYOSTAT PRESSURE BEFORE COMMENCING THIS PROCEDURE.

CAUTION

Replace the Burst Disc immediately after rupturing to avoid cryopumping. Ice blocks can form inside the exhaust lines and Vertical Stack if the Burst Disc is not replaced promptly.

Make sure Coldhead / Shield Cooler power is turned off before starting this procedure to prevent ice formation in Recondenser.

1. Turn off Coldhead / Shield Cooler power.

2-4 BURST DISC REMOVAL (continued)

2. Slowly open the ball valve on the Service Turret to allow any pressure within the Helium Vessel to escape. Do not continue with this procedure until the Pressure Gauges read less than 0.5 psig.
3. Remove the heavy-duty clamp securing the 8 inch (203 mm) Quench Duct (GRP Stub Tube) to the Vent Manifold. See Illustration 2-1.
4. The gasket between the Quench Duct and Vent Manifold is now accessible. The gasket needs to be removed; however, if removal now might damage the gasket, remove it in Step 8.

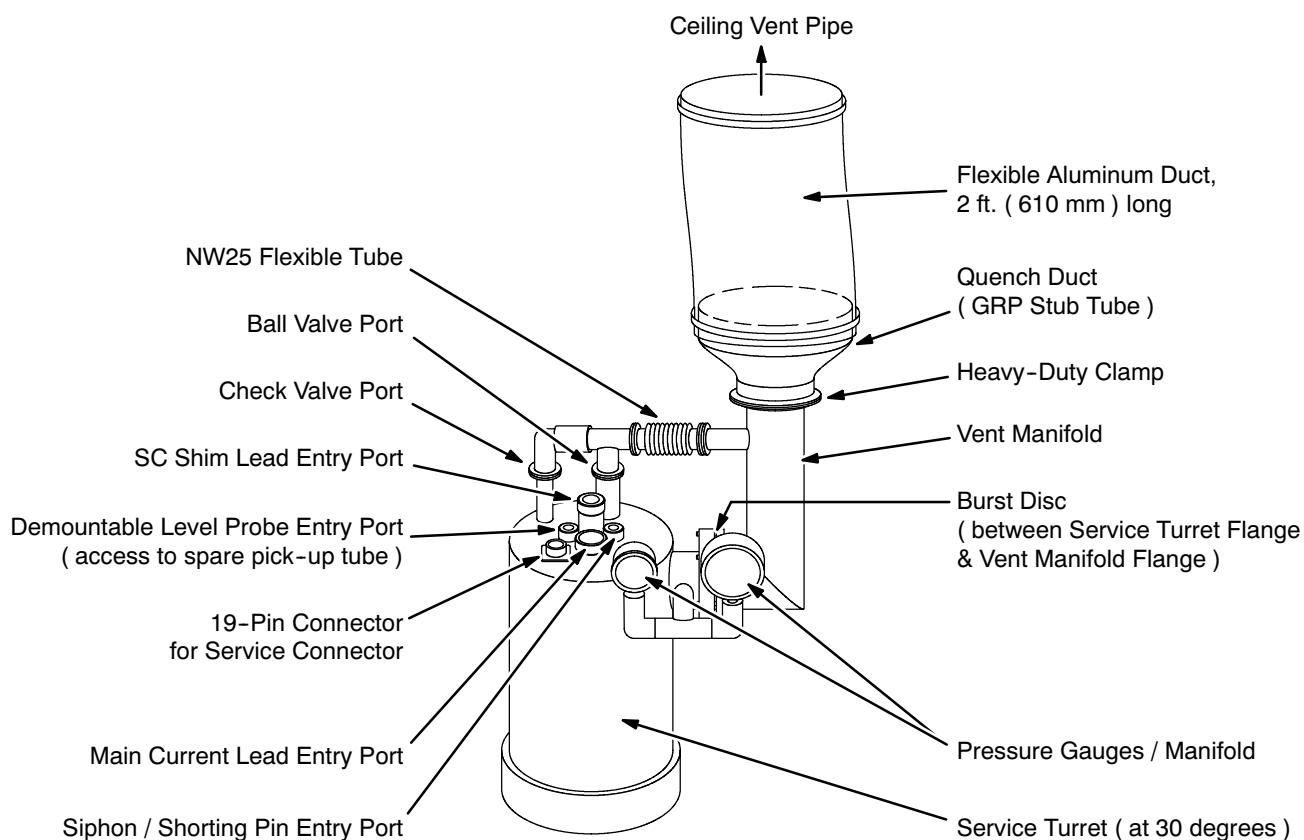
**EXHAUST VENT TO SERVICE TURRET CONNECTIONS**

ILLUSTRATION 2-1

5. Remove the NW25 Clamp securing the Vent Manifold to the NW25 Flexible Tube. See Illustration 2-1.
6. The Centering Ring at the end of the NW25 Flexible Tube is now accessible. The ring needs to be removed; however, if removal now might damage the ring, remove it in Step 8.

2-4 BURST DISC REMOVAL (continued)

7. Using a M5 Allen key, loosen the six M6 cap screws securing the Vent Manifold Flange to the Service Turret Flange. See Illustration 2-1. Do not remove the screws nor allow the Vent Manifold to drop off the Service Turret Flange at this time.
8. Remove the Quench Duct gasket and NW25 Centering Ring if these could not be removed without damage earlier (Steps 3 and 5, respectively).
9. Carefully lower the Vent Manifold to a resting position against the O.V.C.
10. Remove the lowest five M6 cap screws loosened in Step 7, then securely hold the Vent Manifold while removing the remaining M6 cap screw.
11. Take the Vent Manifold off the Service Turret Flange and place the Manifold where it will not fall or get damaged.

Note

The Burst Disc is now accessible for inspection / removal.

2-5 BURST DISC REPLACEMENT

1. Clean any ruptured Burst Disc debris in the exhaust vent system. Make sure the sealing surfaces on the Service Turret Flange and the Vent Manifold Flange are completely clean.
2. Fit a replacement Burst Disc (2255516) onto the Service Turret Flange, with the white Teflon seal in full contact with the flange. See Illustration 2-1. Do not let the Burst Disc fall out.
3. Carefully position the Vent Manifold Flange against the Service Turret Flange while resting the top of the Vent Manifold against the O.V.C.
4. Loosely install six M6 cap screws between the Vent Manifold Flange and the Service Turret Flange. Do not tighten yet.
5. Swing the Vent Manifold into position with the NW25 Flexible Tube. See Illustration 2-1.
6. Install the NW25 Centering Ring between the Vent Manifold and the Flexible Tube, then install the NW25 Clamp.
7. Place the Quench Duct Gasket between the Vent Manifold and the Quench Duct. See Illustration 2-1.
8. Install and tighten the heavy-duty clamp to secure the Quench Duct to the Vent manifold. See Illustration 2-1.
9. Using a M5 Allen key, evenly tighten the six M6 cap screws securing the Vent Manifold Flange to the Service Turret Flange.
10. Close the ball valve on the Service Turret. See Illustration 2-1.

SECTION 3 - LIQUID HELIUM TOP-OFFS



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

During normal magnet operation the helium level in the magnet needs to be maintained above 1864 liters (60% of the vessel's 3066 liter capacity). Consequently, the Helium Level Monitor must be checked regularly. See FUNCTIONAL CHECKS, Section 9, Liquid Helium Level Monitor. A refill should be scheduled whenever the monitor reading from Probe 1 approaches 490 mm. If the monitor's alarm LED is flashing, a refill should be scheduled as soon as possible.

IMPORTANT !!!

Liquid helium is used in this procedure. Make sure of the following prior to starting the precool:

- **The procedure will be performed by trained and authorized personnel.**
- **All required equipment is in place and functional.**

Perform the preparations and follow the procedure found SET-UP AND CALIBRATION Section 9, Liquid Helium Fill / Top-Off, to refill the 3T/940 magnet.

SECTION 4 - RE-EVACUATION OF VACUUM SPACE



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.
- FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.
- HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.

Equipment

- Helium mass-spectrometer leak detector
- Appropriate vacuum hoses

Note

Only use hoses that have not been previously used with gaseous helium.

Procedure

1. Ramp the magnet down to zero field in accordance with REPLACEMENT / MAINTENANCE, Section 1.
2. Connect the leak detector to the vacuum port of _____, but do not open the valve.
3. Evacuate the vacuum hoses to < 10 milliTorr.
4. Set the leak detector to register background level.

Procedure (continued)

5. Slowly open the cryostat valve.
6. Monitor pressure and helium background level.

Note

Helium level may temporarily increase for a minute or so as outgassing causes the valve's o-ring to momentarily unseat.

SECTION 5

SHIELD COOLER COLDHEAD REPLACEMENT



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.
- FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.
- HAVE ALL "WORK ASSISTANTS" OR "WORK OBSERVERS" COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.



THE FOLLOWING SAFETY PRECAUTIONS MUST BE TAKEN BEFORE STARTING TO CHANGE OUT A COLDHEAD OF A MAGNET AT FIELD. THESE PRECAUTIONS ARE REQUIRED TO PREVENT FERROMAGNETIC MATERIAL FROM BECOMING DANGEROUS PROJECTILES IN THE MAGNETIC FIELD OR THE COLDHEAD FROM BEING ATTRACTED INTO THE BORE OR ONTO THE OUTER VESSEL OF THE MAGNET.

- THIS PROCEDURE MUST BE PERFORMED BY PERSONNEL TRAINED AND QUALIFIED IN COLDHEAD REPLACEMENT WHILE THE MAGNET IS AT FIELD.
- NEVER BRING COLDHEAD NEAR THE MAGNET BORE OR IN CONTACT WITH THE OUTER VESSEL DURING REMOVAL / REPLACEMENT.
- DO NOT BRING ANY FERROMAGNETIC TOOLS OR EQUIPMENT INTO MAGNET ROOM. FERROMAGNETIC MATERIAL CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.
- MAKE SURE THE MAGNET RUNDOWN UNIT IS FUNCTIONING PROPERLY TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE FUNCTIONAL CHECKS, SECTION 4.

SECTION 5 - SHIELD COOLER REPLACEMENT (continued)**WARNING!****(CONTINUED)**

- **BOTH OLD AND NEW COLDHEADS MUST BE CARRIED FROM / TO SIDE OF MAGNET IN A DIRECT PATH FROM / TO WALL OF THE ROOM. NEVER BRING THE COLDHEAD NEAR THE END FLANGES OF THE MAGNET WHERE IT CAN BE DRAWN IN THE BORE.**
- **A SECOND PERSON IS REQUIRED TO ASSIST CHANGING OUT A COLDHEAD IN A MOBILE VAN. THIS ENABLES THE PERSON REMOVING THE COLDHEAD TO HAND IT OFF TO THE SECOND PERSON.**
- **IF ALL OF THE ABOVE CONDITIONS ARE NOT MET, THE MAGNET MUST BE RAMPED DOWN BEFORE CHANGING THE COLDHEAD.**

5-1 EQUIPMENT

- 30V 2A Power Supply
- Full helium gas cylinder (99.9995%)
- Regulator Kit (46-306734G1)
- Gas Cylinder Cart (46-258150P1)
- RGD-5100 Coldhead Assembly (2255528)
- High-Vac Pump
- Leak Detector
- 10-Pin Breakout Box (2295140)
- Digital Volt-Ohm Meter (DVOM)
- Various Allen keys, spanners, etc.
- Leather gloves (for opening and closing coldhead motor shield)
- Portable Oxygen Monitor (2106236 or 2106237)
- Flashlight

5-2 REMOVAL

1. Turn the Compressor off.
2. Disconnect the power cable from the Coldhead to the Compressor.



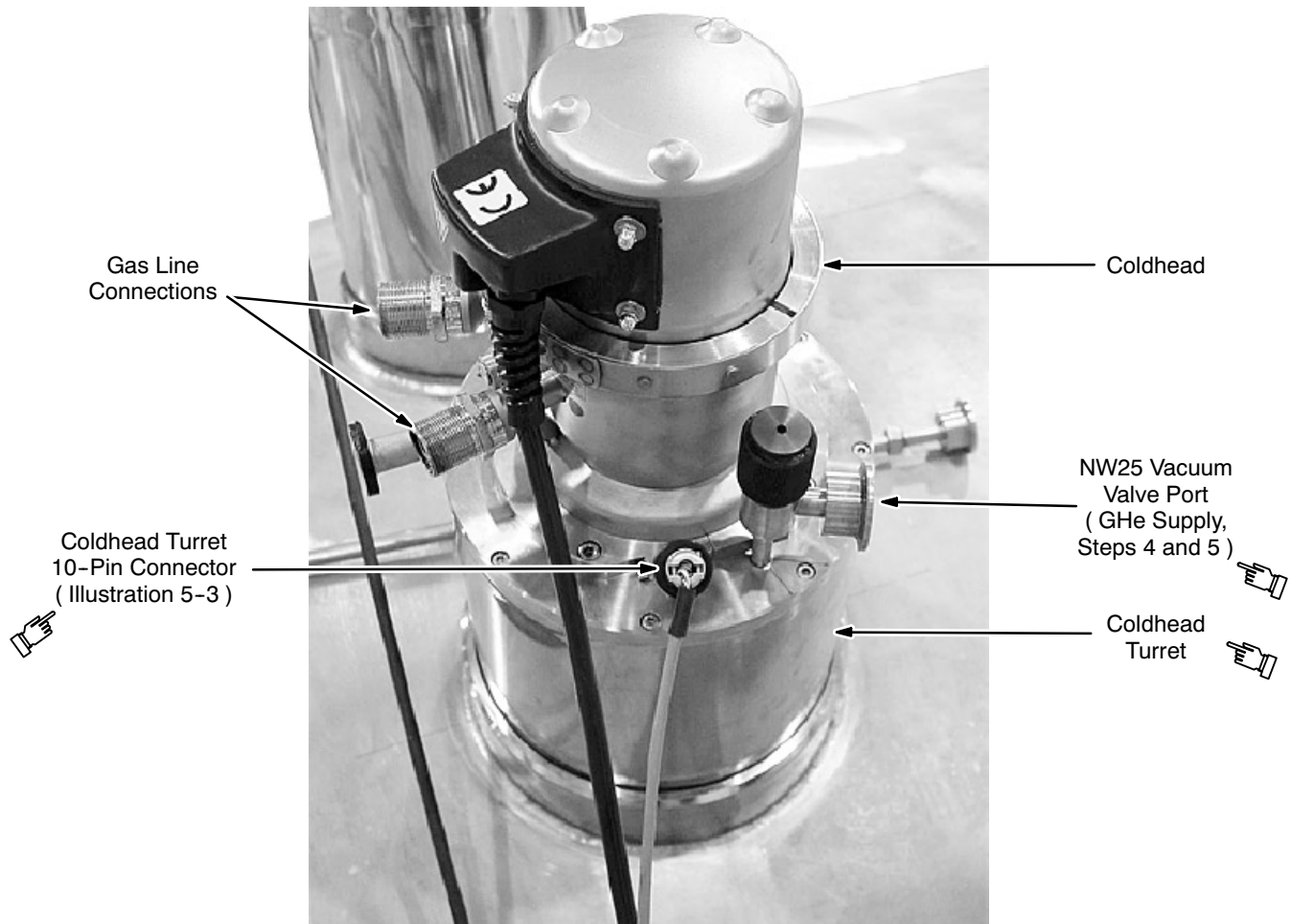
Keep strain relief ty-wraps on gas lines. Do not put a downward force on the Aeroquip fittings during removal of gas line connections to prevent a bending moment on the Aeroquip stem and increase friction force on the Aeroquip fitting threads.

3. Disconnect both Coldhead gas lines at the Coldhead. See REPLACEMENT / MAINTENANCE, Section 11, Helium Flexlines Connections / Replacement.
4. Connect the helium gas supply line to the NW25 Vacuum Valve Port on the Coldhead Turret. Start the gas flowing at 1 – 2 psig.



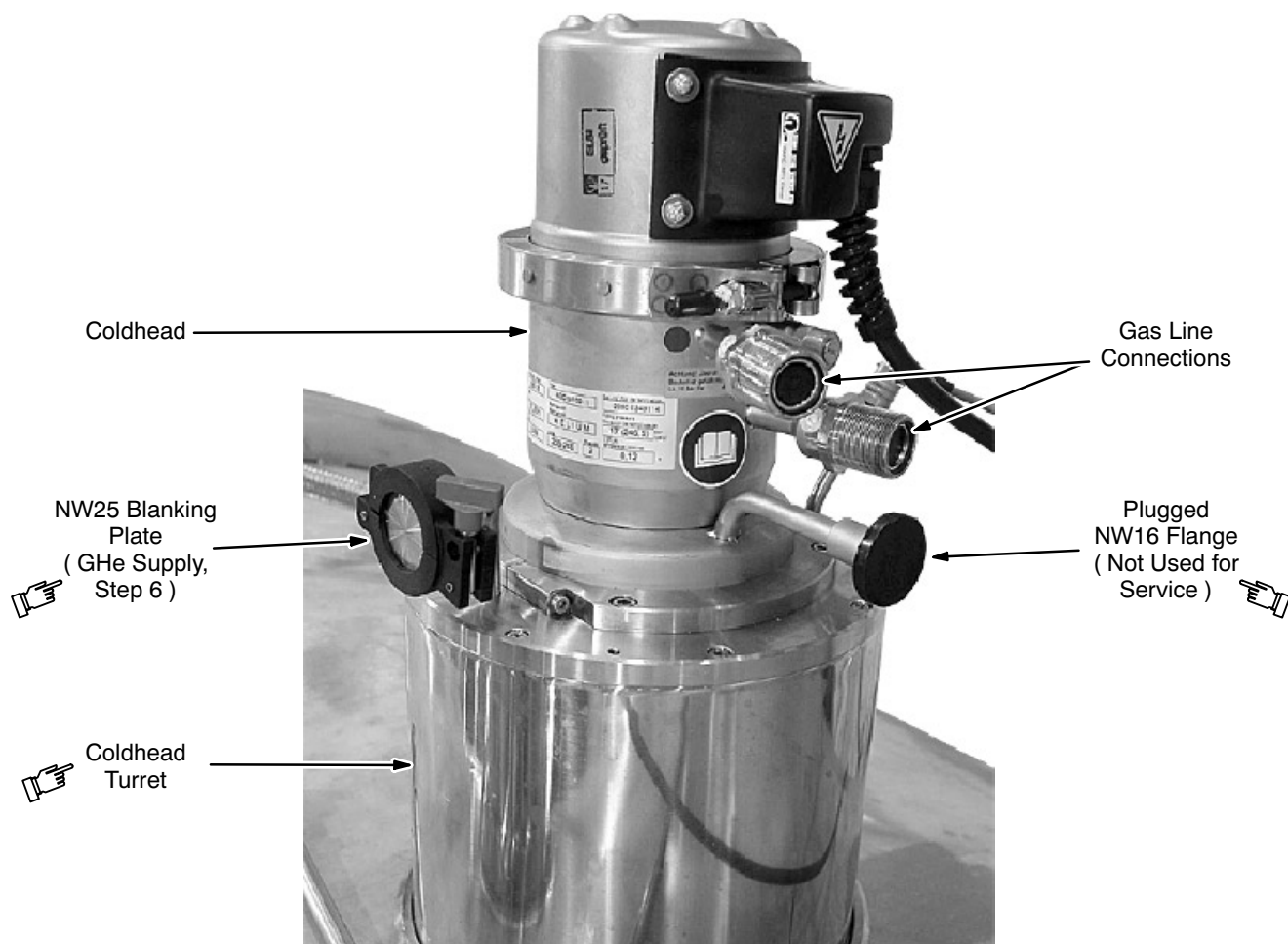
Do not fully open the knob on the NW25 Vacuum Valve Port as a leak can occur while the valve is fully open. Make sure the valve is at least a half turn less than fully open.

5. Vent the vacuum space by allowing helium gas to flow through the NW16 Vacuum Valve on the Coldhead Turret. See Illustration 5-1.

5-2 REMOVAL (continued)

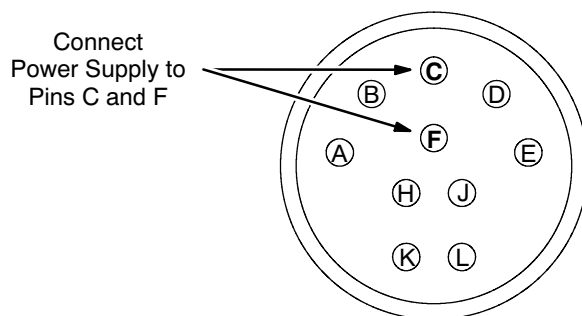
COLDHEAD TURRET AND COLDHEAD
ILLUSTRATION 5-1

6. Remove the NW25 Blanking Plate on the Coldhead Turret. See Illustration 5-2. Allow helium gas to flow through at 1 - 2 psig, measured at the regulator. Make sure this helium gas flow is continuous to keep air / water from entering the turret.

5-2 REMOVAL (continued)

COLDHEAD TURRET AND COLDHEAD
ILLUSTRATION 5-2

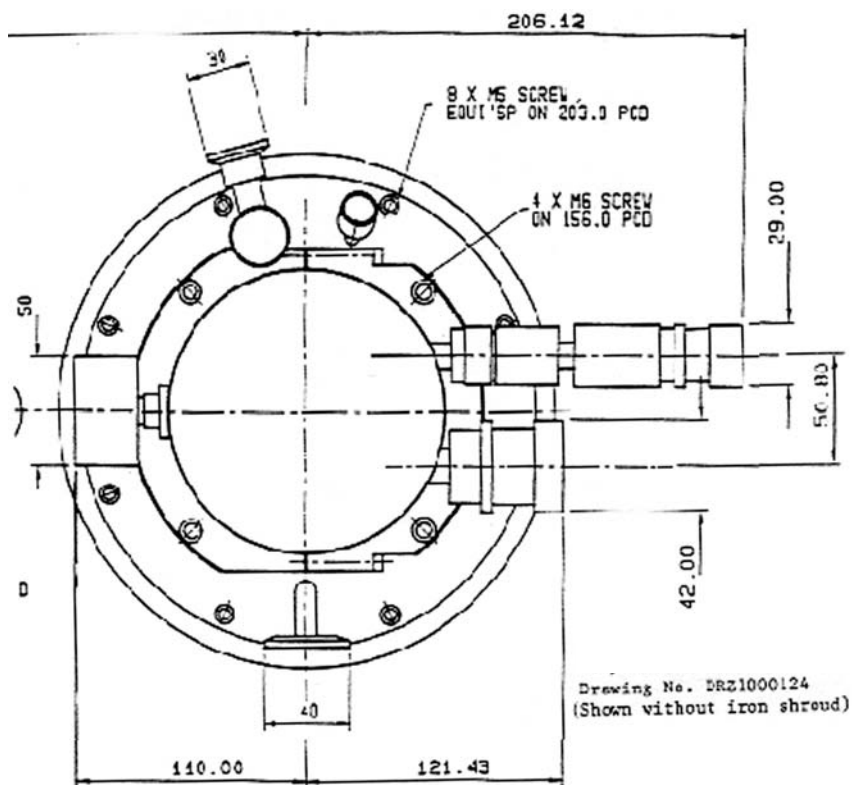
7. Connect the power supply to pins C and F of the Coldhead Turret 10-Pin Connector (Illustration 5-3).



COLDHEAD TURRET 10-PIN CONNECTOR
ILLUSTRATION 5-3

5-2 REMOVAL (continued)

8. Turn the power supply on. Make sure 1 to 1.5 Amps flows to the heater.
9. Monitor the Coldhead first and second stage temperature sensors for warming. The temperature sensors should approach room temperature in approximately 30 minutes. See FUNCTIONAL CHECKS, Section 7, Temperature Sensor Measurement.
10. Once the Coldhead is near room temperature remove the M5 x 8 mm screws from the Coldhead top flange. See Illustration 5-4.



SHIELD COOLER COLDHEAD, TOP VIEW
ILLUSTRATION 5-4

5-2 REMOVAL (continued)**WARNING!**

THE FOLLOWING PRECAUTIONS MUST BE TAKEN WHEN REMOVING THE COLD-HEAD TO PREVENT THE IT FROM BEING ATTRACTED BY THE MAGNETIC FIELD AND AVOID CRYOGEN BURNS:

- **MAKE SURE ALL REQUIRED EQUIPMENT IS ACCESSIBLE WITHIN YOUR REACH.**
- **MAKE SURE TO WEAR NONABSORBENT LEATHER GLOVES WHEN REMOVING THE COLDHEAD. THE FIRST AND SECOND STAGES OF THE COLDHEAD ARE COLD AND CAN CAUSE CRYOGENIC BURNS IF CONTACT IS MADE WITH THE SKIN.**
- **MAKE SURE AN ASSISTANT IS AT THE CRYOSTAT'S SIDE BEFORE DISLODGING COLDHEAD. THE ASSISTANT MUST BE CERTIFIED IN ACCORDANCE WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

11. Try removing the Coldhead by pulling vertically along the Coldhead's axis. If it does not dislodge, wait another 15 minutes before trying again.

WARNING!

DO NOT LOWER OR CARRY COLDHEAD DIRECTLY PAST END OF MAGNET TO PREVENT COLDHEAD BEING DRAWN INTO THE BORE.

12. Immediately pass the dislodged Coldhead to the assistant, who should safely carry the removed Coldhead directly to the wall and then away from the magnet room.

Note

Handle the Coldhead carefully. Do not drop nor allow objects to strike the Coldhead as the lower portion is constructed with thin tubes which can easily be damaged.

13. Quickly cover the turret opening with a blanking flange fully seated on the Coldhead o-ring to prevent air from entering the turret.

- 14. Complete Shield Cooler warming to room temperature.

5-2 REMOVAL (continued)

15. Make sure the Coldhead's copper flanges were not bent during removal.
16. Turn off the heater to pins C - F.
17. Disconnect the power supply.

5-3 REPLACEMENT**WARNING!**

THE FOLLOWING PRECAUTIONS MUST BE TAKEN WHEN REPLACING A COLD-HEAD TO PREVENT THE IT FROM BEING ATTRACTED BY THE MAGNETIC FIELD AND AVOID CRYOGEN BURNS:

- **MAKE SURE ALL REQUIRED EQUIPMENT IS ACCESSIBLE WITHIN YOUR REACH.**
- **MAKE SURE TO WEAR NONABSORBENT LEATHER GLOVES.**
- **USE AN ASSISTANT CERTIFIED IN ACCORDANCE WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

1. Loosen the clamp securing the Shield Cooler to the top flange.
2. Make sure the linkages inside the turret are free of ice. De-ice as needed.

WARNING!

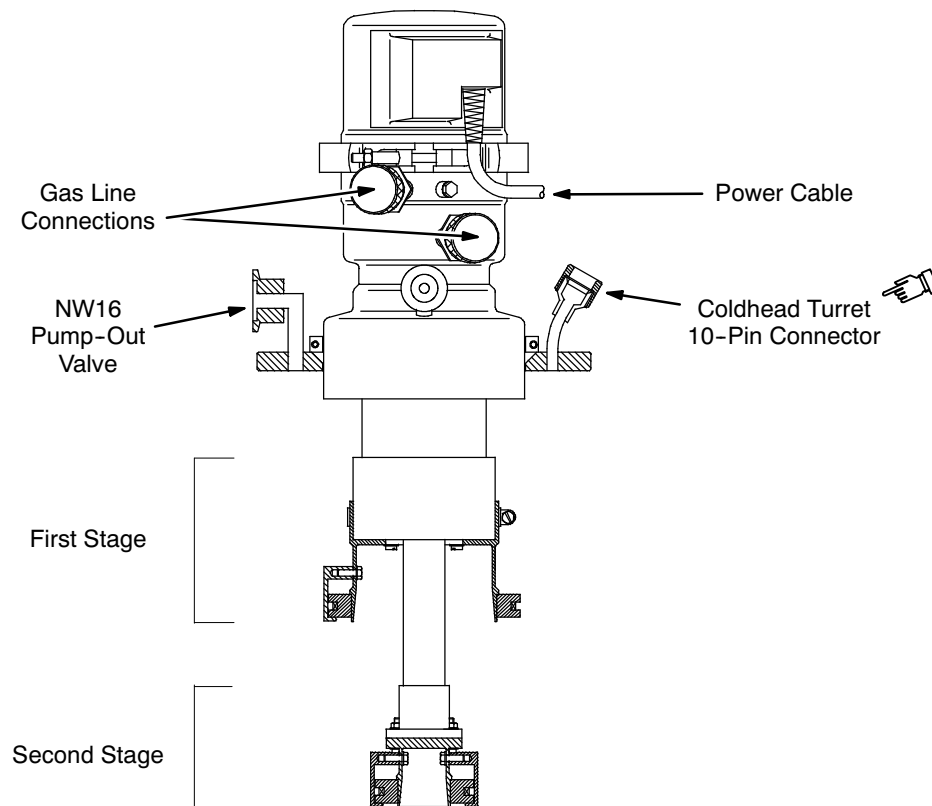
DO NOT CARRY OR RAISE COLDHEAD DIRECTLY PAST END OF MAGNET TO PREVENT COLDHEAD BEING DRAWN INTO THE BORE.

3. Have an assistant carefully bring the replacement Coldhead along the magnet room walls to the magnet's side.

Note

Handle Coldhead carefully. Do not drop nor allow objects to strike Coldhead as lower portion is constructed with thin tubes which can easily be damaged.

4. Insert replacement Coldhead into position in the Cryocooler turret. See Illustration 5-5. Make sure the Coldhead o-ring is fully seated, and the Coldhead is firmly in place.

5-3 REPLACEMENT (continued)

SHIELD COOLER COLDHEAD, SIDE SECTION
ILLUSTRATION 5-5

5. Tighten all Coldhead mounting screws.
6. Replace the NW25 blanking flange on the Coldhead Turret.
7. Repump the vacuum space through the NW16 valve.
8. Tighten the clamp securing the Shield Cooler to the top flange.
9. Check for leaks around the Coldhead o-ring, etc.

Note

The pressure gauge of the Compressor should read 20 -22 bar when running 14 - 16 bas static. A lower reading indicates a leak, which must be corrected by carefully reseating the Coldhead and checking for leaks again.

10. Turn the Shield Cooler on.
11. Monitor the temperature sensors in conformance with FUNCTIONAL CHECKS, Section 7, Temperature Sensor Measurement..

SECTION 6 - DE-ICING MAGNET



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

6-1 INTRODUCTION

Air leaking into the magnet can cause a build-up of ice in the Service Turret's neck that can eventually completely block the exhaust plumbing for helium boil-off. The resulting increased helium vessel pressure can rupture the thin-walled neck tube.

Two gauges on the Service Turret report helium vessel pressure, with the normal pressure range marked in yellow on the right-hand gauge. However, these gauges measure pressure at the top of the neck. They are effected by changes in atmospheric pressure, but will not indicate helium vessel pressure once a complete ice block has developed.

An ice block will be diagnosed when the siphon can not be inserted during helium fill. However, the ice could have been building for some time without obvious symptoms. Once symptoms of ice build-up are apparent, the magnet will require de-icing.



THE MAGNET MUST BE RAMPED DOWN BEFORE BEGINNING DE-ICING PROCEDURE IN ORDER TO PREVENT ANY QUENCH POSSIBILITY. A QUENCH WOULD SUDDENLY RELEASE CRYOGENIC HELIUM GAS INTO THE MAGNET ROOM.

6-1 INTRODUCTION (continued)**IMPORTANT !!!**

Liquid cryogens are released in this procedure. Make sure of the following prior to starting the precool:

- The procedure will be performed by trained and authorized personnel.
- All required equipment is in place and functional.

IMPORTANT !!!

Any ice preventing the insertion / connection of the Ramp Lead must be cleared first to enable the magnet to be ramped down to complete ice clearing. This initial ice clearing will be performed with the magnet at field. The following warnings **MUST** be adhered to prevent exposure to the rapid expulsion of cryogenically cold helium gas and the blow-out tube should the magnet quench during this ice clearing.

WARNING!

THE FOLLOWING SAFETY MEASURES MUST BE TAKEN DURING THIS PROCEDURE TO PREVENT EXPOSURE TO THE RAPID EXPULSION OF CRYOGENIC HELIUM GAS AND BLOW-OUT TUBE SHOULD A MAGNET QUENCH OCCUR:

- **WEAR ALL PERSONAL PROTECTIVE EQUIPMENT REQUIRED FOR CRYOGEN SAFETY (NONABSORBANT GLOVES, JACKET AND PANTS; PROTECTIVE APRON / JACKET; GOGGLES / FACESHIELD).**
- **MAKE SURE THE MAGNET ROOM DOOR IS SECURED IN THE OPEN POSITION AND THAT THE EXHAUST VENT SYSTEM AND FANS ARE OPERATING PROPERLY.**
- **DO NOT PUT YOUR FACE ABOVE OR NEAR THE SERVICE TURRET.**
- **MAKE SURE THIS PROCEDURE IS PERFORMED CAREFULLY BY A PROPERLY TRAINED PERSON.**
- **MAKE SURE THE SERVICE LADDER PLATFORM AT THE MAGNET IS SECURE AND ALLOWS RAPID ESCAPE.**
- **MAKE SURE AN EXIT PATH AT LEAST 28 INCHES (711 MM) WIDE IS MAINTAINED AT ALL TIMES.**
- **MAKE SURE A SECOND PERSON TRAINED IN MAGNET / CRYOGEN SAFETY PRACTICES AND FAMILIAR WITH THE SITE (EXITS, OUTSIDE PHONE LOCATION, ETC.) IS PRESENT.**
- **UNDERSTAND AND COMPLY WITH ALL MAGNET SAFETY REQUIREMENTS AND SERVICE SAFETY REQUIREMENTS ITEMS PUBLISHED FOR GE SUPERCONDUCTING MAGNETS.**
- **IF A QUENCH STARTS TO OCCUR, ABANDON ALL WORK UNDERWAY AND EXIT THE ROOM IMMEDIATELY.**

6-2 TOOLS / EQUIPMENT

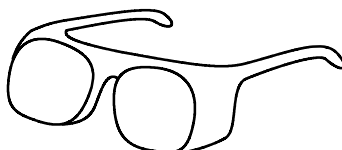
6-2-1 Personal Protection Equipment

- Goggles, face shield and gloves from the Cryogen Safety Kit 46-271137G1. See Illustration 6-1.
- Protective Apron / Jacket.
- Portable Oxygen Monitor (2106236 or 2106237) if Oxygen Monitor Kit 2107184 is not installed and functional in room.
- Cryogen Warning Signs specified in 2301164PRE, *MR Magnet - Safety Requirements*.



Face
Shield*

Leather
Gloves*



Goggles*



Protective
Apron / Jacket

* Contained in Cryogen Safety Kit (46-271137G1).

SAFETY EQUIPMENT

ILLUSTRATION 6-1

6-2-2 Tools

- Blow-Out Tube (2255509; for 2.9M ceiling: 2294703) with Phase Separator (2294434)
- Source of pure helium gas (Dewar or gas bottle)
- Connecting hose. If a ferromagnetic helium gas bottle is used, use a 15 meter (50 foot) nonferrous extension hose so the bottle may be located outside the magnet's 10 gauss zone.

6-3 PREPARATION

WARNING!

- **HELIUM GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**
 - A. MAKE SURE THAT THE MAGNET ROOM DOOR IS PROPPED OPEN,**
 - B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND**
 - C. THE PORTABLE FLOOR FAN IS INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THIS PROCEDURE.**
- **MAKE SURE ALL REQUIRED PERSONNEL PROTECTIVE EQUIPMENT IS WORN TO PREVENT CRYOGEN EXPOSURE / BURNS: GOGGLES / FACE SHIELD, NONABSORBENT GLOVES, LONG-SLEEVE SHIRT AND LONG PANTS, PROTECTIVE APRON / JACKET.**
- **SMOKING IS PROHIBITED AROUND CRYOGENS. CRYOGENS CAN CONDENSE ATMOSPHERIC OXYGEN PRODUCING A HIGHLY COMBUSTION SUPPORTING FLUID. CLOTHING WHICH HAS ABSORBED LIQUID OXYGEN CAN REMAIN FLAMMABLE FOR LONG PERIODS OF TIME.**
- **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT PROPERLY FUNCTIONING OXYGEN MONITORS ARE WORN ON ALL PERSONNEL.**

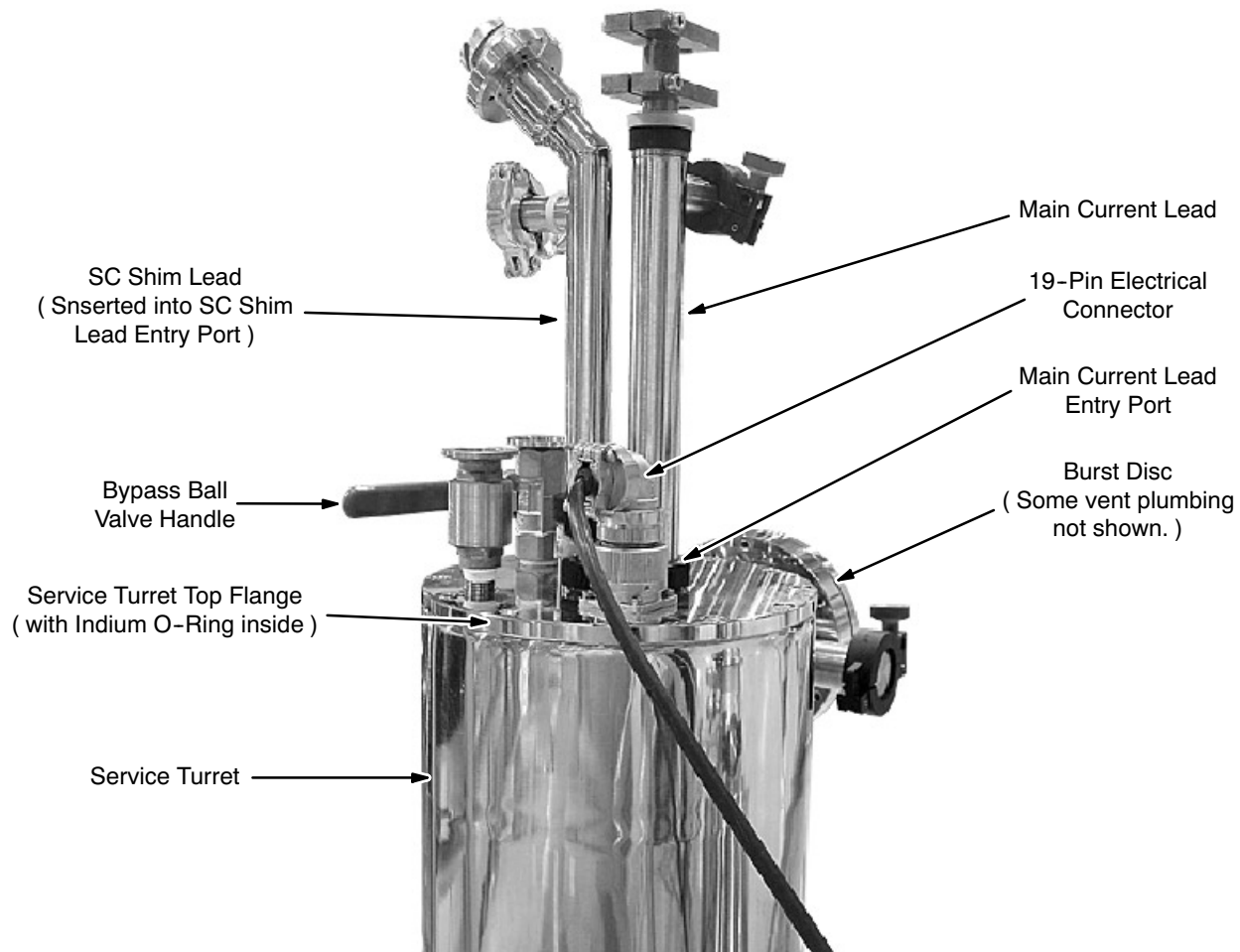
1. Complete all items marked "X" in the "Magnet / Recondenser De-Icing" column of SAFETY REQUIREMENTS, 2301164PRE, MR Magnet - Safety Requirements, Table 1, Service Procedure Safety Requirements.

IMPORTANT !!!

Make sure all identified De-Icing defects are corrected before continuing with magnet service.

6-4 PROCEDURE

1. Open the blue-handled Bypass Valve on the Service Turret by moving the handle to the vertical position. See Illustration 6-1.



SERVICE TURRET
ILLUSTRATION 6-1

2. Continue venting until the sound of escaping gas is no longer heard, indicating the upper neck is at atmospheric pressure.

Note

No sound indicates EITHER a) the helium vessel is already at atmospheric pressure or b) the exhaust passage is completely blocked by ice.

3. Connect the long hose from the helium gas bottle to the end of the Blow-Out Tube. Make sure the Phase Separator is in place on the Blow-Out Tube's end.

6-4 PROCEDURE (continued)

4. Start the helium gas flowing at 1.0 psig.
5. Carefully unscrew the Siphon Entry Port's cap and lift the plug.

WARNING!

WHEN THE BLOW-OUT TUBE BREAKS THROUGH AN ICE BLOCKAGE A SUDDEN EXPULSION OF CRYOGENIC HELIUM GAS MAY OCCUR.

6. Slowly insert the end of the Blow-Out Tube into the Siphon / Shorting Pin Entry Port until resistance from ice is felt.
7. Continue pushing the Blow-Out Tube into the ice with a small force while the helium gas begins melting the ice. If after 3 minutes the Blow-Out Tube has not pushed further into the ice, increase the helium flow to 2.0 psig.
8. If the ice block can not be cleared in 30 minutes, contact / call your "MAC" Team Leader.

CAUTION

Do not allow large amounts of warm helium gas into the lower neck as a magnet quench may result.

9. When the Blow-Out Tube has broken through the ice, retract it 0.5 inch (10 mm) or so.
10. Open the Ball Valve Port to allow excess gas to vent from the helium vessel into the magnet exhaust vent.
11. Once pressure has been dissipated, direct the Blow-Out Tube to flow warm helium gas on the remaining ice.
12. Make sure ice is clear from the Shorting Pin connector:
 - a. When ice has cleared from the Shorting Pin, remove the pin with the Shorting Pin Insertion Tool (2255522).
 - b. Screw the Blow-Out Tube into the Shorting Pin connector at least two turns.
 - c. Flow 2 psig helium gas until the bottom connector is clear of ice.
 - d. Remove the Blow-Out Tube.

6-4 PROCEDURE (continued)

13. Replace the Siphon Port plug and nut once the port is clear of ice.
14. Stop the helium gas flow.
15. Repeat Steps 4 - 14 to clear ice from the SC Shim Lead Entry Port and the Main Current Lead Entry Port.

IMPORTANT !!!

At Step 12 make sure to test insert the SC Shim Lead in its connector and the Main Current Lead in its connector. If connection isn't made, flow 2 psig helium gas on the connector until obstructing ice is cleared.

16. Diagnose the source of the ice formation:
 - a. Verify that the Burst Disc has not ruptured and is seated correctly. Replace or reseal the Burst Disc as necessary.
 - b. Check the o-rings in the Siphon Entry Port, the SC Shim Lead Entry Port and the Main Current Lead Entry Port for breakage. Check the nuts for the ability to tighten. Replace o-rings / nuts and reseal the port plugs as necessary.
 - c. Make sure the Top Flange is seated and that the indium seal is not leaking. Tighten screws as necessary.
 - d. Make sure the Service Turret Check Valve and Ball Valve seals do not leak. Replace the seals using Teflon or PTFE tape as necessary.
 - e. Verify that all Service Turret and Vent Kit fittings are correctly assembled and installed.

SECTION 7 - MAGNET WARM-UP



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

7-1 INTRODUCTION

Very few situations will require warming up the magnet in the field (for example, removing an internal ice block or field repairing a vacuum leak that requires reevacuation). If a magnet warm-up is required, contact the Regional MAC Team Representative before proceeding.

IMPORTANT !!!

Liquid cryogens are released in this procedure. Make sure of the following prior to starting the precool:

- **The procedure will be performed by trained and authorized personnel.**
- **All required equipment is in place and functional.**

7-2 TOOLS / EQUIPMENT

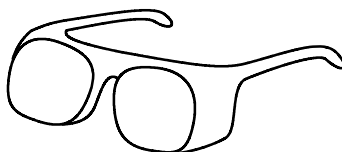
7-2-1 Personal Protection Equipment

- Goggles, face shield and gloves from the Cryogen Safety Kit 46-271137G1. See Illustration 7-1.
- Protective Apron / Jacket.
- Portable Oxygen Monitor (2106236 or 2106237) if Oxygen Monitor Kit 2107184 is not installed and functional in room.
- Floor exhaust fan. Fan to be specified by Installation Specialist and obtained locally.
- Cryogen Warning Signs specified in *2301164PRE, MR Magnet - Safety Requirements*.



Face
Shield*

Leather
Gloves*



Goggles*

Protective
Apron / Jacket



* Contained in Cryogen Safety Kit (46-271137G1).

SAFETY EQUIPMENT

ILLUSTRATION 7-1

7-2-2 Tools

- Empty 1500 liter helium Dewars to hold recovered liquid helium
- Full helium gas cylinder with pressure regulator
- Two full dry nitrogen gas cylinders
- 0.5 inch (12.7 mm) diameter Helium Transfer Siphon (2255866; for 2.9M ceilings: 2292330)
- Burst Disc Blank
- Pump-Out Tool (See Illustration 7-2.)
- Pressure gauge
- Vacuum fittings (o-rings and clamps)
- NW40 to NW25 reducer
- Diaphragm valve
- Non-return check valve
- NW25 tee

7-3 PREPARATION

WARNING!

- **HELIUM GAS GENERATED DURING THIS PROCEDURE IS ODORLESS, COLORLESS AND DISPLACES OXYGEN IN THE AIR. TO PREVENT ASPHYXIATION:**
 - A. MAKE SURE THE MAGNET ROOM DOOR IS PROPPED OPEN,**
 - B. THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS AND**
 - C. THE PORTABLE FLOOR FAN IS INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THIS PROCEDURE.**
 - **MAKE SURE ALL REQUIRED PERSONAL PROTECTIVE EQUIPMENT IS WORN TO PREVENT CRYOGEN EXPOSURE / BURNS: GOGGLES / SAFETY GLASSES AND FACE SHIELD; INSULATED NONABSORBENT GLOVES, LONG-SLEEVE SHIRT AND LONG PANTS; AND PROTECTIVE APRON / JACKET.**
 - **SMOKING IS PROHIBITED AROUND CRYOGENS. CRYOGENS CAN CONDENSE ATMOSPHERIC OXYGEN PRODUCING A HIGHLY COMBUSTION SUPPORTING FLUID. CLOTHING WHICH HAS ABSORBED LIQUID OXYGEN CAN REMAIN FLAMMABLE FOR LONG PERIODS OF TIME.**
 - **MAKE SURE A FUNCTIONAL OXYGEN MONITOR HAS BEEN INSTALLED IN THE MAGNET ROOM IN CONFORMANCE WITH THE SITE REQUIREMENTS OR THAT PROPERLY FUNCTIONING OXYGEN MONITORS ARE WORN ON ALL PERSONNEL.**
 - **METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD. DO NOT BRING ANY FERROMAGNETIC OBJECTS, EQUIPMENT OR TOOLS INTO THE MAGNET ROOM.**
1. Complete all items marked "X" in the "Helium Precool / Warm-Up" column of SAFETY REQUIREMENTS, *2301164PRE, MR Magnet - Safety Requirements*, Table 1, Service Procedure Safety Requirements.

IMPORTANT !!!

Make sure all identified Helium Precool / Warm-Up defects are corrected before continuing with magnet service.

7-4 PROCEDURE**WARNING!**

DO NOT BRING A FERROMAGNETIC HELIUM GAS BOTTLE INSIDE MAGNET FIELD'S 10 GAUSS ZONE. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.

1. Ramp the magnet down in accordance with REPLACEMENT / MAINTENANCE Section 1, Rampdown to Zero Field.

WARNING!

Do not activate the Magnet Rundown Unit for this rampdown.

2. Remove liquid helium from the cryostat:
 - a. Fully screw the M10 thread of the Helium Transfer Siphon into the magnet's siphon cone.
 - b. Exchange the Burst Disc with a Burst Disc Blank.
 - c. Pressurize the helium vessel to 4.5 psig.
 - d. Insert other end of transfer siphon in empty helium Dewar when liquid helium is seen emerging from siphon end.
3. When helium has been removed from cryostat, vent helium gas through Cryocooler "sock."

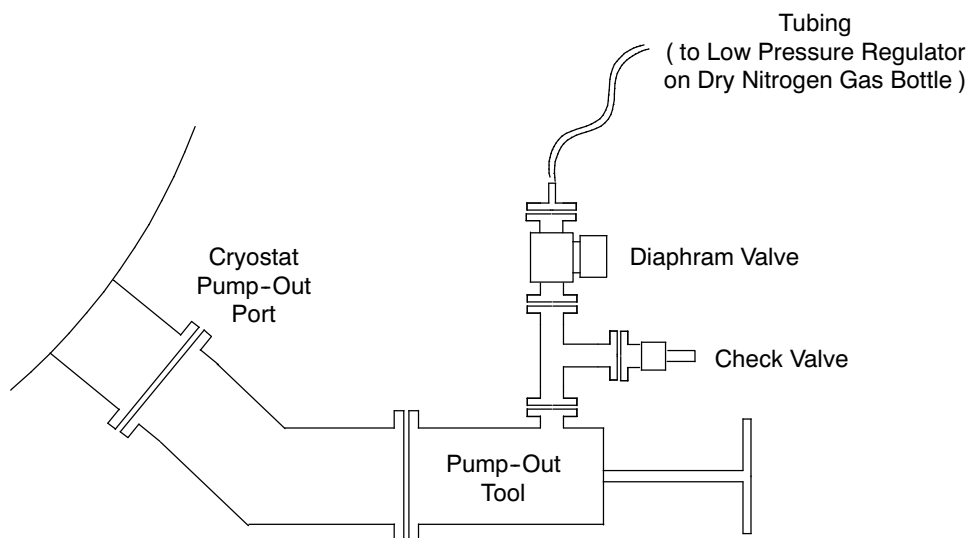
Note

Vent the Cryocooler "sock" before venting the main vacuum to prevent possible "sock" damage.

7-4 PROCEDURE (continued)

Remove the Room Temperature Shim Tube before venting the Cryostat vacuum in Step 4 to prevent the tube being crushed.

4. Vent the cryostat vacuum vessel with dry nitrogen gas:
 - a. Connect the Pump-Out Tool to the cryostat's Pump-Out Port. See Illustration 7-2.
 - b. Attach a check valve, diaphragm valve, tubing and bottle of dry nitrogen gas with a low pressure regulator to the Pump-Out Tool as shown in Illustration 7-2.



PLUMBING ARRANGEMENT FOR CRYOSTAT VENTING

ILLUSTRATION 7-2

- c. Make sure the low pressure regulator is working by operating it and then back its valve fully out.
 - d. Screw the Pump-Out Tool plunger into the tapped hole in the brass transit bung.
 - e. Pull the Pump-Out Tool plunger out very slowly.
 - f. Open the valve on the bottle of dry nitrogen gas and slowly twist in the regulator valve to start gas flow through this plumbing and into the cryostat.
 - g. Continue the gas flow until pressure within the vacuum space is slightly above atmospheric pressure. This may take one or two hours.
 - h. Use the gas bottle's regulator to balance pressure against the check valve to keep pressure within the vacuum space at 0.0 - 0.5 psig.
5. Leave cryostat in this condition for at least seven days without leakage.

SECTION 8 - TRANSITING THE MAGNET

8-1 INTRODUCTION

The magnet should not be moved without the Transit Bungs installed to secure the internal cryostat vessel.

IMPORTANT !!!

The magnet must first be warmed up in accordance with **REPLACEMENT / MAINTENANCE** Section 7 before commencing this procedure.

8-2 PROCEDURE



The outer cover plates are heavy. To avoid injury:

- Wear leather gloves.
- Use each cover plate's top bolt for support while removing the other eleven bolts, then hold the plate with a "free" hand while removing the top bolt. Use both hands when lowering each plate.

IMPORTANT !!!

Remove one pair of outer and inner cryostat cover plates and reinstall one transit bung assembly at a time.

1. Remove the eleven lowest M12 bolts securing one outer cover plate in place. Only slightly loosen the top bolt. See Illustration 7-3.
2. Remove the top bolt while securely holding the cover plate with a "free" hand. Carefully lower the plate with both hands. Rest the plate with the o-ring surface facing up.

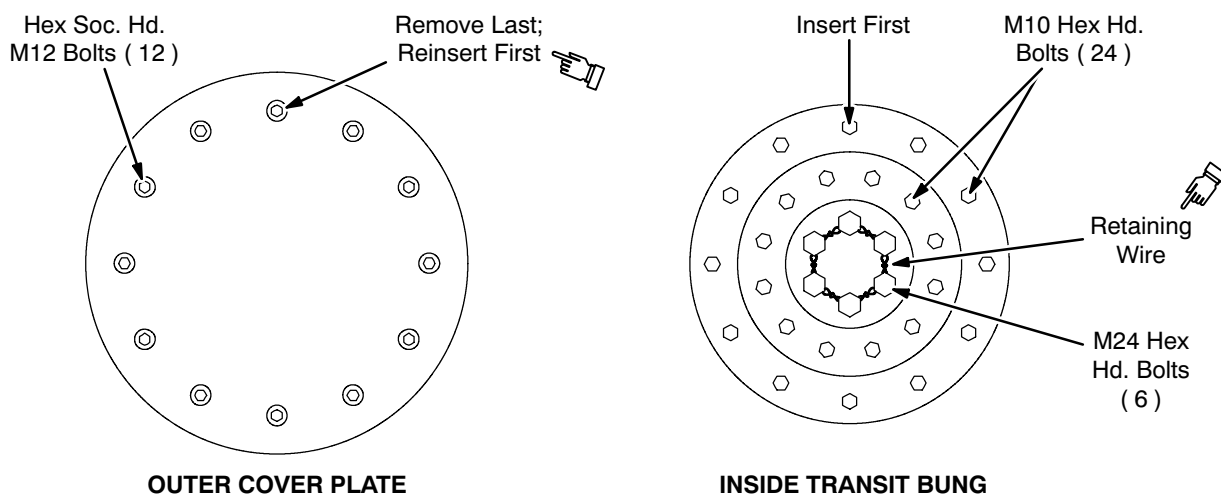
8-2 PROCEDURE (continued)**BUNG BOLT ARRANGEMENTS**

ILLUSTRATION 7-3

3. Remove the larger inner thermal vessel cover plate. Set plate and fasteners aside.

IMPORTANT !!!

Make sure no component part or tool is dropped inside the cryostat as they can not be retrieved.

4. Remove the mylar superinsulation patches from the plate.
5. Remove the smaller inner thermal vessel cover plate. Place plate and fasteners with the larger cover plate.

CAUTION

The transit bung weighs approximately 35 pounds (15.88 kg). To avoid injury:

- **Wear leather gloves.**
- **Have a second technician use both hands to lift each bung and hold in place while the top bolt is inserted and secured.**

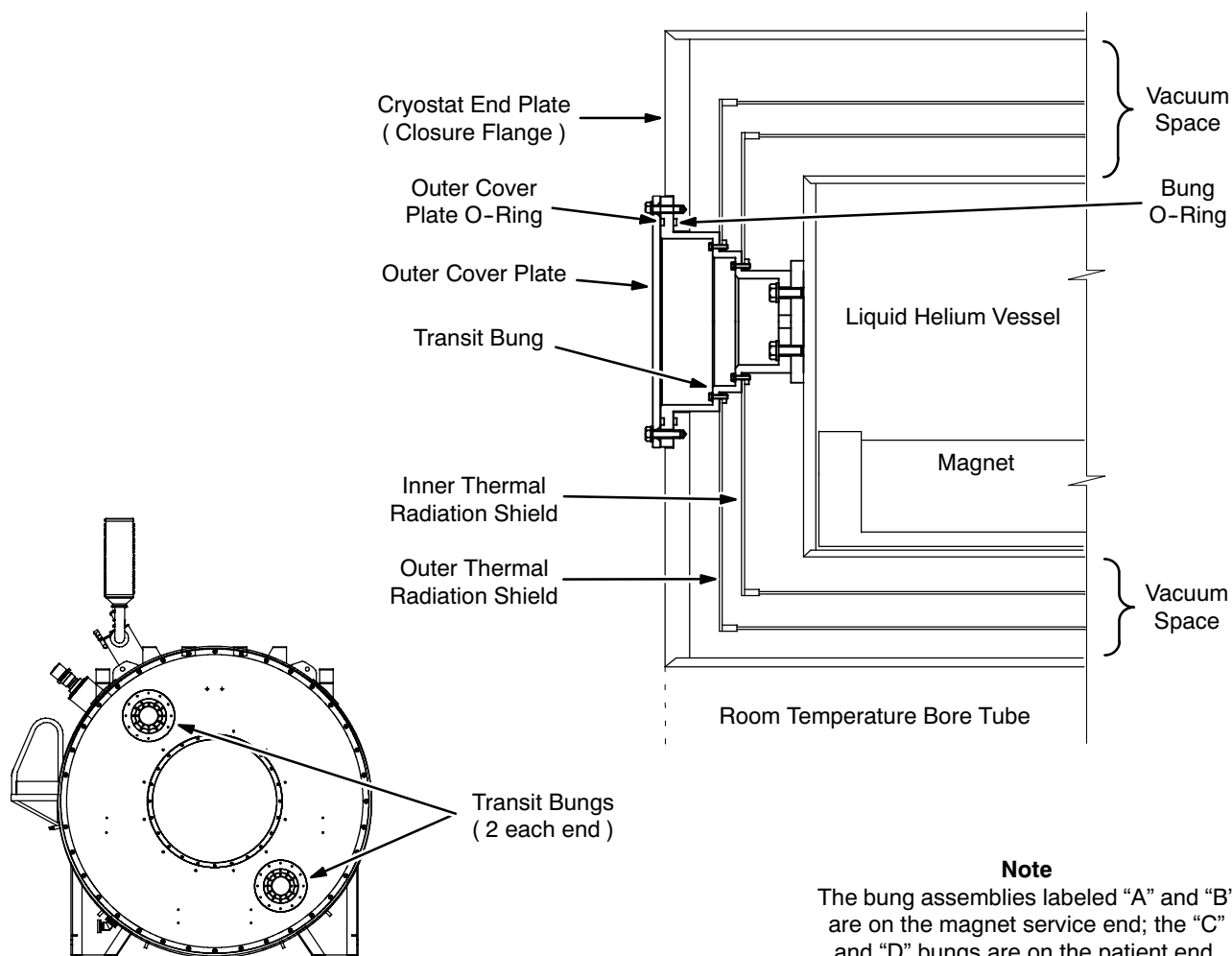
6. Clean with isopropyl alcohol and lightly grease the Transit Bung o-ring.
7. Fit the Transit Bung o-ring into the o-ring groove on the cryostat end plate.

8-2 PROCEDURE (continued)

8. Have a second technician wearing leather gloves carefully lift the magnet's Transit Bung Assembly into the bung hole and align the bung bolt holes with those of the vessel. See Illustration 7-4.
9. Insert the top M10 bolt and secure sufficiently to hold the bung's weight. See Illustration 7-3.
10. Insert the six M24 bolts into the innermost holes in the bung. Only loosely tighten. See Illustration 7-3.
11. Insert the other 23 M10 bolts into the Transit Bung Assembly. Only loosely tighten. See Illustration 7-3.
12. Temporarily insert four M12 bolts in opposing holes to locate the Bung Assembly with the cryostat end plate. Only loosely tighten. See Illustration 7-3.

Note

At this point the bung should be fully located.

**Note**

The bung assemblies labeled "A" and "B" are on the magnet service end; the "C" and "D" bungs are on the patient end.

TRANSIT BUNGS
ILLUSTRATION 7-4

8-2 PROCEDURE (continued)

13. Fully tighten all M10 and M24 bolts in a cross-pattern sequence. Do not tighten the M12 bolts. See Illustration 7-3.
14. Remove the four M12 bolts loosely inserted in Step 12.
15. Clean the original outer cover plate o-ring with isopropyl alcohol and lightly grease.
16. Mount the outer cover plate o-ring in the o-ring groove of the bung. See Illustration 7-4.



The outer cover plates are heavy. To avoid injury:

- **Wear leather gloves.**
- **Use each cover plate's top bolt for support while removing the other eleven bolts, then hold the plate with a "free" hand while removing the top bolt. Use both hands when lowering each plate.**

17. Replace the outer cover plate on the cryostat end plate and secure with all twelve M12 bolts.
18. Evacuate the cryostat vacuum vessel to 0.1 mb in accordance with REPLACEMENT / MAINTENANCE, Section 4, Re-Evacuation Of Vacuum Space.

The magnet is now ready to be shipped.

SECTION 9 - QUENCH RECOVERY

Introduction

During a magnet quench extremely cold helium gas ruptures the Burst Disc to escape through the Exhaust Vent System, making the Service Turret very cold and allowing air into the helium vessel. The helium vessel can begin cryopumping, and enough ice can form in the Service Turret to make Main Current Lead and SC Shim Lead insertion impossible. The magnet must be returned to its operating condition quickly to minimize cryopumping and icing. All debris and ice must be cleared prior to helium fill.

Procedure



DO NOT ENTER THE MAGNET ROOM UNTIL THE ROOM / PERSONAL OXY-GEN MONITOR INDICATEDS ROOM CONDITIONS ARE SAFE.



THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RECOVERING FROM A MAGNET QUENCH:

- **MAKE SURE THE MAGNET ROOM DOOR IS PROPPED OPEN.**
- **THE MAGNET ROOM EXHAUST SYSTEM AND VENT SYSTEM ARE INSTALLED AND FUNCTIONING IN CONFORMANCE WITH THE SITE REQUIREMENTS.**
- **MAKE SURE A PROPERLY FUNCTIONING OXYGEN MONITOR IS USED IN THE MAGNET ROOM BY SERVICE PERSONNEL.**
- **REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, INCLUDED WITH THIS MANUAL.**

IMPORTANT !!!

If a magnet quench occurs, perform the following procedure immediately in order to minimize “cryopumping” and icing in the magnet.

Report all magnet quenches via The Online Quench Report.

Procedure (continued)

1. Verify with a functioning Oxygen Monitor that the magnet room is safe to enter.
2. Allow the helium exhaust from the magnet to subside. This typically takes at least 30 minutes.
3. Replace the Burst Disc in conformance with REPLACEMENT / MAINTENANCE, Section 2.
4. Deice the Service Turret in conformance with REPLACEMENT / MAINTENANCE, Section 6.

IMPORTANT !!!

Make sure the Shorting Pin, SC Shim Lead and Main Current Lead connectors are free of ice.

5. Order liquid helium as quickly as possible and replace in conformance with SET-UP AND CALIBRATION, Section 9, Liquid Helium Fill, as soon as liquid helium is on-site.

Note

If liquid helium is not immediately available, make sure that a positive cryostat pressure is maintained. If necessary, pressurize the cryostat with 99.999% helium gas until liquid helium arrives.

6. Inspect the Vent System for damage and debris. Clear out debris and repair damage as required.
7. Check the plumbing and pressure gauges for leaks and / or damage. Repair and replace components as necessary.
8. Order a replacement Burst Disc (2255516) to keep on-site as a spare.
9. Make sure the magnet is filled to the 100% level with liquid helium. Check magnet pressure.
10. Ramp and shim the magnet in conformance with SET-UP AND CALIBRATION, Sections 10 and 11, respectively.

SECTION 10 - PREPARATION FOR SITE REMOVAL

1. Perform the following procedures from this manual (Direction 2295400) in sequence:
 - a. Ramp down the magnet in conformance with REPLACEMENT / MAINTENANCE, Section 1, Ramp-down to Zero Field.
 - b. Warm up the magnet in conformance with REPLACEMENT / MAINTENANCE, Section 7, Magnet Warm-Up.
 - c. Prepare the magnet for transit in conformance with REPLACEMENT / MAINTENANCE, Section 8, Transiting the Magnet.
2. Move the magnet in conformance with Direction 2341078, GEMSO 3.0T/940 Active Shield Magnet Delivery and Installation, Section 5, Moving Magnet to MR Suite.

SECTION 11

HELIUM FLEXLINE CONNECTIONS / REPLACEMENT

11-1 INTRODUCTION

Follow the procedures in this section to prevent helium loss and damage to the Coldhead and Aeroquip fittings during gas line replacement. Make sure the gas line insulation is reinstalled on the new gas lines to prevent “spike” noise and image problems.



Order and technique are important when connecting helium flexlines to prevent inoperative Coldheads and system contamination.



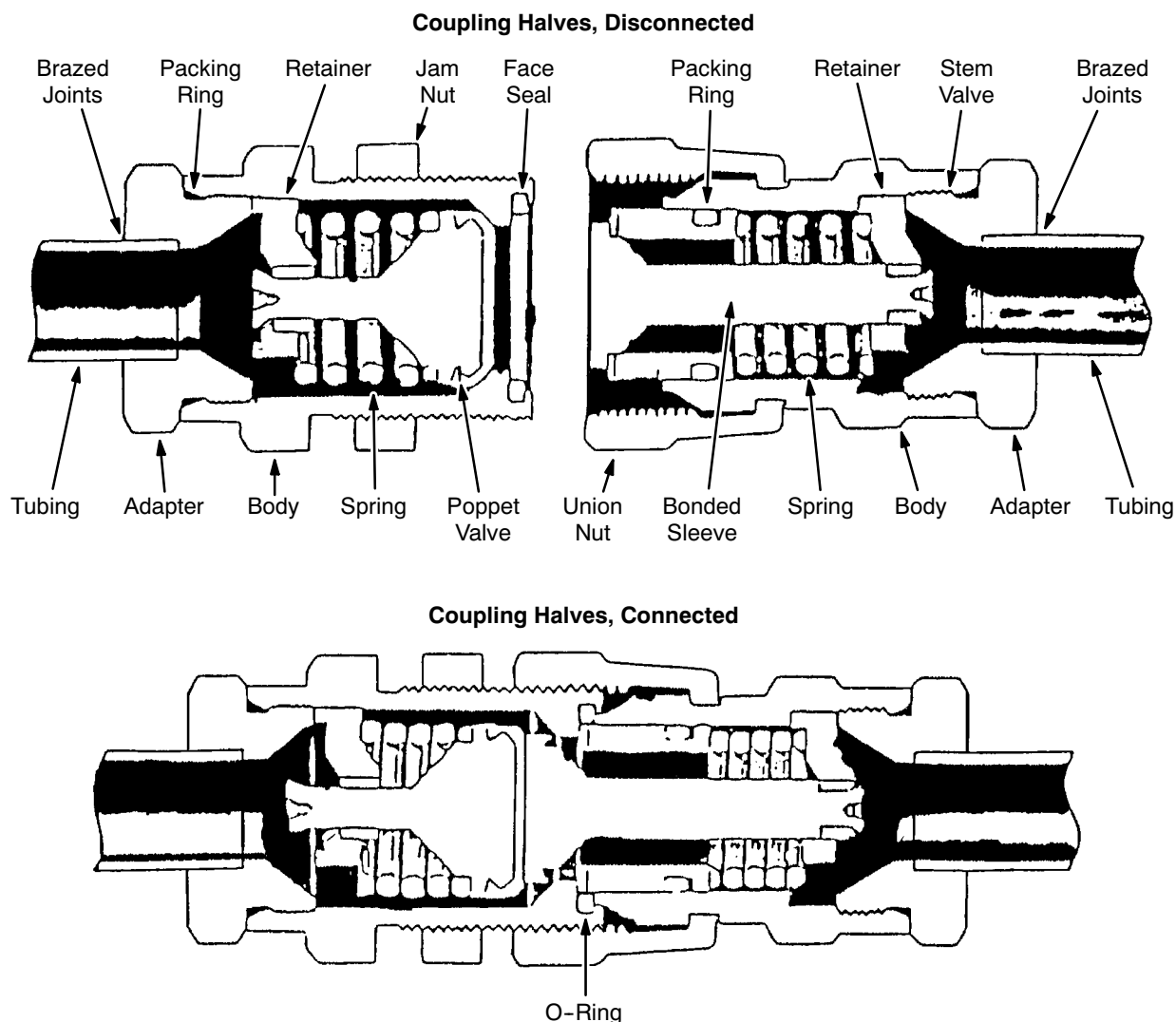
DO NOT PUT ANY BENDING FORCE ON AEROQUIP FITTINGS WHILE CONNECTING / DISCONNECTING HELIUM FLEXLINES. A BENDING FORCE WILL CREATE DIFFICULTY IN THE RAPID THREAD ENGAGEMENT / DISENGAGEMENT REQUIRED TO PREVENT HELIUM LOSS AND CONTAMINATION. IT CAN ALSO RESULT IN BENDING OR LEAKING OF THE AEROQUIP STEMS. SUPPORT THE GAS LINES TO PREVENT A BENDING FORCE WHEN CONNECTING / DISCONNECTING TO THE COLDHEAD AT THE TOP OF THE MAGNET OR TO THE COMPRESSOR.

11-2 TOOLS / EQUIPMENT

- Aeroquip fitting wrenches (included in Cold Head / Compressor Installation & Maintenance Kit 46-281088G3)
- Flexible Corrugated Plastic Raceway Tubing 2251611
- Clean, lint-free nonabrasive cloth

11-3 CONNECTING AEROQUIP COUPLINGS**Note**

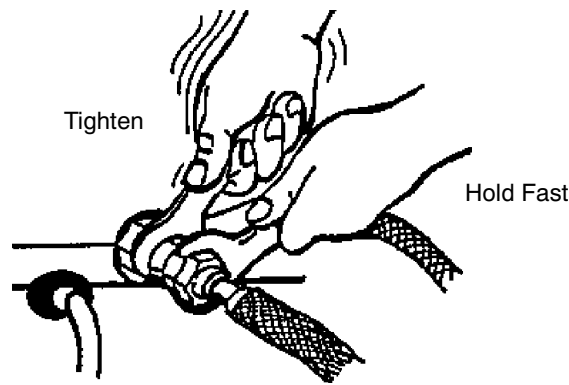
Refer to Illustrations 11-1 through 11-3 for this procedure. Make sure the Aeroquip o-ring is present and in good condition on female fittings on the Coldhead side.



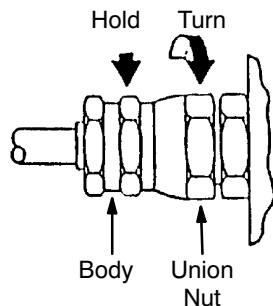
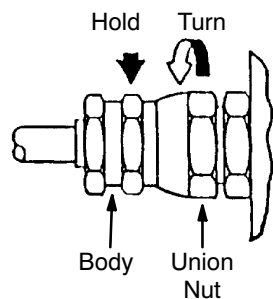
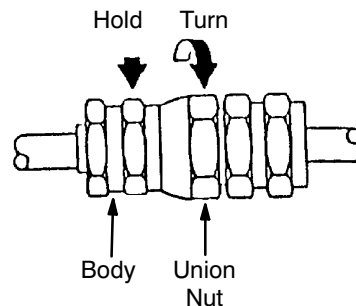
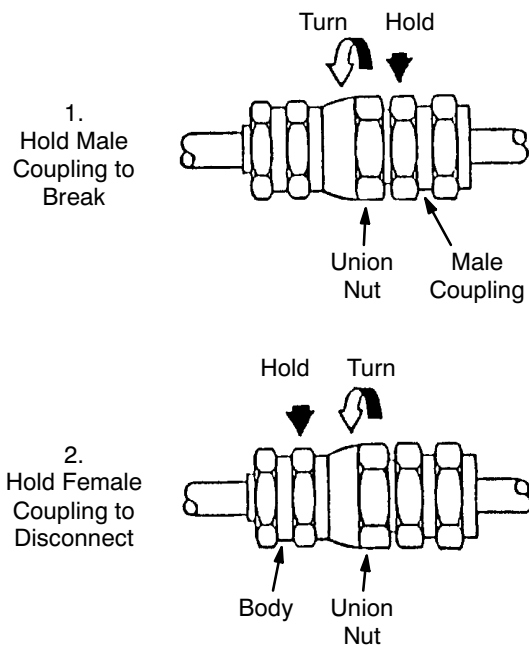
VIEW OF DISCONNECTED AND CONNECTED SELF-SEALING (AEROQUIP) COUPLINGS
ILLUSTRATION 11-1

11-3 CONNECTING AEROQUIP COUPLINGS (continued)

1. The system gas connections are shipped with caps and plugs to keep the fittings clean and free from damage. Remove the caps and plugs and thread them together for storage.
2. Wipe the faces of the couplings with a lint-free cloth to insure they are clean and free of chips and dust.
3. Make sure the face seal is in place on the inside periphery of the male coupling and is not damaged.
4. To make the connection, start the hose-side union nut onto the male connector by hand. Then, with the wrenches supplied, hold the stationary part of the female coupling while turning the union nut with the other wrench.
5. As the poppet begins to open there will be a slight venting of gas from the fitting. Continue to tighten the connection until the female coupling is firmly seated against the face seal on the male coupling. The required torques are:
 - 35 ft-lbs (47.5 N-m) for a 1/2 inch connection
 - 45 ft-lbs (61.0 N-m) for a 3/4 inch connection



PROPER 2-WRENCH CONNECTION TECHNIQUE FOR AEROQUIP COUPLINGS
ILLUSTRATION 11-2

11-3 CONNECTING AEROQUIP COUPLINGS (continued)**Connecting to Compressor****Disconnecting from Compressor****Connecting to Coldhead****Disconnecting from Coldhead**

CONNECTING AND DISCONNECTING GAS LINES
ILLUSTRATION 11-3

11-4 DISCONNECTING AEROQUIP COUPLINGS**IMPORTANT !!!**

Make sure the male couplings at the Compressor and Coldhead DO NOT ROTATE when disconnecting lines. Avoid torsional forces on the flex sections.

Note

Excessive gas will escape if the fittings are not aligned properly during connection or disconnection.

1. To disconnect the gas line at the Coldhead, first use one wrench to turn the female coupling union nut about 1/8 turn, while holding the male coupling with the other wrench. This will overcome the initial torque required to break the connection without loosening the male connector from its adapter.
2. Make sure the hose does not rotate to avoid a torsional force on the hose.
3. Place the second wrench on the stationary part of the female coupling and continue to unthread the union nut. Make sure the male connector does not rotate when disconnecting.
4. Make sure the bulkhead jam nut is secure and the male coupling does not rotate when removing the gas line.
5. To disconnect the gas line at the Compressor, turn the union nut on the female coupling while holding the stationary part of female coupling with a second wrench. Since the male coupling is bulkheaded to the Compressor front panel with a lock washer, the male coupling should not rotate from its adapter while removing.
6. When the hoses are disconnected, check each coupling to make sure the face seal is in place.

Note

While the hose is venting during disconnection, the face seal is often blown out of its gland and into the female coupling. Failure to remove the seal from the female coupling will cause the connection to leak when reconnected, with or without another face seal installed.

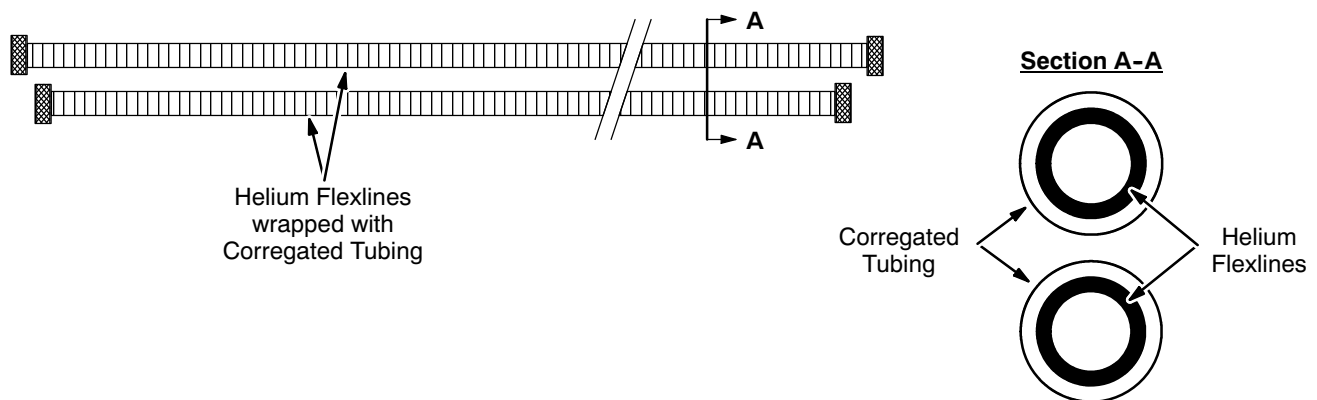
7. If all seals are in place, replace the dust caps and plugs to the coupling halves.

11-5 REPLACING GAS LINE INSULATION CASING

1. Slide the old insulated tubing off the gas line. See Illustration 11-4.
2. Cover both lines completely with Corrugated Plastic Tubing (2251611) from the Penetration Panel where the lines exit the magnet room to the helium line fittings at the Coldhead. See Illustration 11-4.

Note

Corrugated plastic tubing is used to isolate helium gas lines from each other and from ground to prevent spike noise.



HELIUM LINE CASING INSULATION
ILLUSTRATION 11-4

SECTION 12 - PERIODIC MAINTENANCE

Maintenance procedures should be performed at the intervals specified in Table 10-1. Some procedures require specially trained personnel.

TABLE 10-1
MAINTENANCE INTERVALS

Item	Perform Once Every				Trained Service Personnel Required?
	Work Day	Two Months	Year	Two Years	
1. Check for presence of spare parts kit and service ladder for magnet.	X				
2. Inspect magnet for ice (REPLACEMENT / MAINTENANCE Section 6, De-Icing Magnet)	X				
3. Inspect for damage at vertical stack and plumbing joints.	X				
4. Measure cryogen level (FUNCTIONAL CHECKS Section 9, Liquid Helium Level Measurement)	X				
5. Calculate helium boil-off rate (FUNCTIONAL CHECKS Section 10, Cryogen Boil-Off Rate Calculation)	X				
6. Check helium vessel operating pressure (FUNCTIONAL CHECKS Section 8, Cryostat Pressure Measurement)	X				
7. Check Coldhead operation (46-294439P4, Leybold Coldhead vendor manual)	X				
8. Check Magnet Rundown Unit (FUNCTIONAL CHECKS Section 4, Magnet Rundown Unit)	X				
9. Check Shield Cooler Compressor operation (FUNCTIONAL CHECKS Section 5, Shield Cooler Compressor Checks)		X			
10. Measure Cryostat Shield temperature (FUNCTIONAL CHECKS Section 7, Temperature Sensor Calibration & Measurement)		X			
11. Check Compressor cooling water flow rate (46-294439P4, Leybold Coldhead vendor manual)		X			
12. Replace Coldhead Displacer (46-294439P4, Leybold Coldhead vendor manual)			X		Yes
13. Replace Burst Disc (REPLACEMENT / MAINTENANCE Section 2, Burst Disc Replacement)			X		Yes
14. Replace Shield Cooler Compressor Absorber (46-294439P4, Leybold Coldhead vendor manual)				X	Yes

